



Calculus on extensional fuzzy numbers with applications

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ABSTRACT

This article introduces calculus concepts on the set of extensional fuzzy numbers and illustrates their novel contributions to investigations about fuzzy differential equations. At first, we equip the set of extensional fuzzy numbers with a metric to build it into complete metric spaces. Then, we introduce and explore basic properties of the limit, continuity, derivative, and integral of functions, taking values as extensional fuzzy numbers. Especially, the integral is studied under both the Riemann sum and Lebesgue's measure approaches. The novel calculus is built based on the arithmetics of extensional fuzzy numbers introduced in [1] in which the mentioned concepts for extensional fuzzy numbers are a natural extension of corresponding concepts on real numbers. Consequently, it inherits most of properties of the original that provide efficient tools for studies on fuzzy differential equations. In addition to several examples illustrating the calculation with the proposed concepts, we demonstrate valuable contributions of the proposed calculus to the theory and application of fuzzy differential equations, more precisely the fuzzy initial value problems.

1. Introduction

There is no doubt that fuzzy set theory is one of the most widely applied mathematical theories since it was published by Lotfi A. Zadeh in the 1960s in [2]. The success of this theory comes from its ability to model vague information of phenomena that is unavoidable when building models based on experimental data or human intuition, which are always uncertain and personal. Most fuzzy models used in practice are built based on fuzzy logic, fuzzy set theory, or "higher-level" theories, such as the theory of fuzzy integrals, fuzzy differential (integral) equations, or fuzzy fractional (partial) differential equations. These theories are generalizations of classical theories where the extensions of a (crisp) set to a fuzzy set and a real number to a fuzzy number play central roles. However, it is a fact that the research and application of the mentioned (higher-level) fuzzy theories are still limited compared to the corresponding theories in classical mathematics. The essential reason is that their foundation (fuzzy calculus) fails to maintain the natural properties of the calculus with real numbers that bring many advantages in building fundamental concepts such as the limit, derivative, or integral.

Recall that fuzzy calculus, especially the arithmetic operations on fuzzy numbers such as addition, subtraction, multiplication, and division, has attracted the attention of researchers very soon since the 1970s, mostly based on Zadeh's extension principal and standard arithmetic operations of intervals [3–5]. These approaches do not preserve the existence of inverse elements concerning addition and multiplication. It is a shortcoming to define a metric based on subtraction and absolute function, as in the case of real numbers. As a result, to introduce fundamental concepts as the limit or derivative of fuzzy-valued functions, one must base on the

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so-called Hausdorff distance and generalized subtraction, i.e., Hukuhara difference, generalized Hukuhara difference [6–8]. This leads to some crucial properties relating to standard arithmetic operations being complicated or no longer guaranteed. To overcome these shortcomings, researchers have proposed various approaches for generalizing the standard arithmetic operations of real numbers to the case of fuzzy numbers, for example, constrained fuzzy arithmetic [9,10], fuzzy arithmetic operations using gradual numbers [11], or fuzzy arithmetic based on multidimensional representation of fuzzy numbers [12]. These approaches can retain most of the properties of standard arithmetic operations in the case of fuzzy numbers. However, they commonly have complicated computations.

The paper [1] seems to be a significant difference from the earlier mentioned works. It is based on a different way of looking at fuzzy numbers (a fuzzy number is a fuzzy point on the real line). It is known that fuzzy number is the underlying concept of studies on fuzzy calculus. The works mentioned in the previous paragraph consider fuzzy numbers as a generalization of sets (particularly real intervals) in the real line. More precisely, a fuzzy number is a fuzzy set on the real line, fulfilling specific assumptions that accept its core as an interval. This leads to some properties of standard arithmetic operations being lost in fuzzy cases since calculating with real numbers and calculating with sets are fundamentally different in classical mathematics. For the present investigation, we agree that fuzzy numbers are fuzzy points [13,14] that are natural extensions of real numbers (points) on the real line. We restrict our study to the extensional fuzzy numbers defined in [1]. This kind of fuzzy number is an elegant interpretation of the vagueness of real numbers, e.g., “about 3.5”, “approximately 5”, and so on. An extensional fuzzy number is defined by a real number and a fuzzy similarity relation, which characterizes a vague region around the real point. The arithmetic operations on extensional fuzzy numbers are defined in [1] followed by two phases: applying standard operations to the core and maximizing the fuzziness of fuzzy numbers. This approach breaks away from traditional methods, sticking to the computations with α -cuts of fuzzy numbers or Zadeh’s extension principle. It retains the intuitive nature of classical arithmetic with fuzzy numbers, simplifies the calculation, and preserves essential algebraic properties that are valid with real numbers.

Based on the convenience of the arithmetic operations on the extensional fuzzy numbers mentioned above, we propose calculus concepts for the set of fuzzy numbers of this type and the class of functions defined on a real interval and valued in this set. The proposed concepts are constructed in the spirit of the definitions of arithmetic operations, namely acting on the core and the fuzziness of extensional fuzzy numbers. We equip the set of extensional fuzzy numbers with a metric that combines the standard metric in the real line and the metric between fuzzy similarity relations. This space establishes the foundation for building the concepts and exploring the properties of the limit, continuity, derivative, and integral of functions, valued as extensional fuzzy numbers. We show that these concepts inherit most of the properties of the corresponding concepts on the set of real numbers, including a generalization of the chain rule and a formulation of the mean value theorem in a fuzzy setting. Especially, we introduce two approaches for defining the integral of a fuzzy-valued function: the first is based on the traditional Riemann sum, and the second is more general based on the Lebesgue’s approach. In addition to several examples illustrating the calculation with the proposed concepts, the last but very motivating part is devoted to novel contributions of the proposed calculus to the theory and application of fuzzy differential equations, especially the fuzzy initial value problem. The most interesting result is that a fuzzy initial value problem may have infinitely many solutions with the same core but differing in the fuzziness they describe. We demonstrate that this interesting thing opens up the possibility of using fuzzy differential equations in practice in which one can select a solution that adapts with given data, rather than a rigid solution that may be far from the data of previous methods. A practical model with a fuzzy initial value problem is analyzed to illustrate the efficiency of the proposed approach.

This article is structured as follows. Section 2 provides basic knowledge about fuzzy numbers and specific classes of fuzzy similarity relations. The central part of this article investigating the calculus on extensional fuzzy numbers is presented in Section 3. Section 4 presents an application of the proposed calculus to the initial value problem, including numerical examples and comparisons with some benchmark approaches. The last section is to conclude.

2. Basic notions

It is known that a fuzzy set on a universe U is characterized by a function from U to interval $[0, 1]$. This function is called the membership function of the fuzzy set. A fuzzy number is a specific fuzzy set on $\mathbb{R}$. Various definitions of a fuzzy number have been proposed since Lotfi A. Zadeh first introduced it. Let us recall the most often used one below.

Definition 1. A fuzzy set $\mathcal{A}$ on $\mathbb{R}$ is said to be a fuzzy number, provided that

- i) there is $x \in \mathbb{R}$ such that $\mathcal{A}(x) = 1$,
- ii) for any $\lambda \in [0, 1]$, $x, y \in \mathbb{R}$,

$$\mathcal{A}(\lambda x + (1 - \lambda)y) \geq \min\{\mathcal{A}(x), \mathcal{A}(y)\},$$
- iii) $\mathcal{A}$ is upper semicontinuous on $\mathbb{R}$,
- iv) the closure of $\{x \in \mathbb{R} \mid \mathcal{A}(x) > 0\}$ is compact.

$\mathcal{A}$ is called a *continuous fuzzy number* if it is continuous on $\mathbb{R}$.

A fuzzy similarity relation on $\mathbb{R}$ is a special fuzzy set on $\mathbb{R} \times \mathbb{R}$.

Definition 2. [1] A function $\mathcal{E} : \mathbb{R} \times \mathbb{R} \rightarrow [0, 1]$ is said to be a *fuzzy similarity relation* (or *fuzzy similarity*, for short) concerning a continuous t-norm $\otimes$ if the following statements hold:

- i) $\mathcal{E}(x, x) = 1$, for any $x \in \mathbb{R}$,

- ii) $\mathcal{E}(x, y) = \mathcal{E}(y, x)$, for any $x, y \in \mathbb{R}$,
- iii) $\mathcal{E}(x, y) \otimes \mathcal{E}(y, z) \leq \mathcal{E}(x, z)$, for any $x, y, z \in \mathbb{R}$.

Note that the t-norm $\otimes$ in the previous definition can be a weaker form, namely, a left-continuous t-norm. For the theory proposed in our works, we assume that $\otimes$ is continuous. Moreover, the following additional assumptions on a fuzzy similarity need for introducing the arithmetics on extensional fuzzy numbers; see [1].

- iv) If $\mathcal{E}(x, y) = 1$, then $x = y$,
- v) $\mathcal{E}(x + z, y + z) = \mathcal{E}(x, y)$, for any $x, y, z \in \mathbb{R}$,
- vi) $\mathcal{E}(a, \cdot)$ is a continuous fuzzy number, for any $a \in \mathbb{R}$.

In what follows, we recall two classes of fuzzy similarities often used in the theory and application of extensional fuzzy numbers.

Example 1. The function $\mathcal{E}_{\beta,L} : \mathbb{R} \times \mathbb{R} \rightarrow [0, 1]$ with $\beta \in (0, \infty)$, defined by

$$\mathcal{E}_{\beta,L}(x, y) = \max\{1 - \beta|x - y|, 0\},$$

is a fuzzy similarity with respect to the Łukasiewics t-norm. Similarly, the function $\mathcal{E}_{\beta,P} : \mathbb{R} \times \mathbb{R} \rightarrow [0, 1]$ with $\beta \in (0, \infty)$, defined by

$$\mathcal{E}_{\beta,P}(x, y) = e^{-\beta|x-y|},$$

is a fuzzy similarity with respect to the product t-norm. It is easy to see that these fuzzy similarities fulfill all statements from i) to vi) mentioned above.

In the present paper, for any set $I \subset (0, \infty]$, denote $\mathbb{E}_L^I = \{\mathcal{E}_{\beta,L} \mid \beta \in I\}$ and $\mathbb{E}_P^I = \{\mathcal{E}_{\beta,P} \mid \beta \in I\}$, where $\mathcal{E}_{\infty,L}$ and $\mathcal{E}_{\infty,P}$ are defined as the crisp equality, i.e.,

$$\mathcal{E}_{\infty,L}(x, y) = \mathcal{E}_{\infty,P}(x, y) = \begin{cases} 1, & x = y \\ 0, & x \neq y \end{cases},$$

for any $x, y \in \mathbb{R}$.

Definition 3. A continuous fuzzy number Γ is said to be a *S-generation* concerning a t-norm $\otimes$ if the following statements hold

- i) $\Gamma(x) = \Gamma(-x)$, for any $x \in \mathbb{R}$,
- ii) $\Gamma(x) = 1$ iff $x = 0$,
- iii) $\Gamma(x) \otimes \Gamma(y) \leq \Gamma(x + y)$, for any $x, y \in \mathbb{R}$.

Lemma 1. Let $\mathcal{E}$ be a fuzzy similarity with respect to a t-norm $\otimes$. Let $\Gamma : \mathbb{R} \rightarrow [0, 1]$ be defined by

$$\Gamma(x) = \mathcal{E}(0, x), \quad \text{for any } x \in \mathbb{R}. \quad (1)$$

Then, Γ is a *S-generation* concerning t-norm $\otimes$.

Proof. The proof is straightforward from the properties of a fuzzy similarity. $\square$

Definition 4. Let $\mathcal{E}$ be a fuzzy similarity and Γ be a function defined as in Lemma 1. We call Γ the *S-generation* of $\mathcal{E}$.

It follows from this definition that if Γ is the S-generation of $\mathcal{E}$ then $\mathcal{E}$ is determined from Γ as follows

$$\mathcal{E}(x, y) = \Gamma(x - y), \quad \text{for any } x, y \in \mathbb{R}. \quad (2)$$

In this case, we say that $\mathcal{E}$ is generated by Γ .

Lemma 2. Let $\mathbb{E}$ be a set of fuzzy similarities and $\mathbb{G}$ be the set of S-generations of fuzzy similarities in $\mathbb{E}$. Let $\psi : \mathbb{E} \rightarrow \mathbb{G}$ be defined by $\psi(\mathcal{E}) = \Gamma$, for any $\mathcal{E} \in \mathbb{E}$ and Γ is the S-generation of $\mathcal{E}$. Then, ψ is a bijective mapping.

Proof. It is easy to see that $\psi(\mathcal{E}) \in \mathbb{G}$ for any $\mathcal{E} \in \mathbb{E}$. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathbb{E}$, and Γ_1, Γ_2 be their S-generations, respectively. Assume that $\mathcal{E}_1 = \mathcal{E}_2$, we obtain $\psi(\mathcal{E}_1)(x) = \Gamma_1(x) = \mathcal{E}_1(0, x) = \mathcal{E}_2(0, x) = \Gamma_2(x) = \psi(\mathcal{E}_2)(x)$, for any $x \in \mathbb{R}$. It follows that $\psi(\mathcal{E}_1) = \psi(\mathcal{E}_2)$. Now, assume that $\psi(\mathcal{E}_1) = \psi(\mathcal{E}_2)$. It means that $\Gamma_1 = \Gamma_2$. Therefore, $\Gamma_1(x) = \Gamma_2(x)$, for any $x \in \mathbb{R}$. Since, $\mathcal{E}_1(x, y) = \mathcal{E}_1(0, x - y) = \Gamma_1(x - y)$ and $\mathcal{E}_2(x, y) = \mathcal{E}_2(0, x - y) = \Gamma_2(x - y)$, we obtain that $\mathcal{E}_1(x, y) = \mathcal{E}_2(x, y)$, for any $x, y \in \mathbb{R}$. Consequently, $\mathcal{E}_1 = \mathcal{E}_2$. Finally, let $\Gamma \in \mathbb{G}$ be the S-generation of $\mathcal{E} \in \mathbb{E}$. It is clear that $\psi(\mathcal{E}) = \Gamma$. Therefore, ψ is a bijective mapping. $\square$

It follows from this lemma that a set of fuzzy similarities is isomorphic to the set of S-generation of fuzzy similarities in it. Particularly, a fuzzy similarity is uniquely determined by an S-generation and vice versa.

Example 2. The function $\Gamma_{\beta,L} : \mathbb{R} \rightarrow [0, 1]$ with $\beta \in (0, \infty)$, defined by

$$\Gamma_{\beta,L}(x) = \max\{1 - \beta|x|, 0\},$$

is the S-generation of the fuzzy similarity $\mathcal{E}_{\beta,L}$. The function $\Gamma_{\beta,P} : \mathbb{R} \rightarrow [0, 1]$ with $\beta \in (0, \infty)$, defined by

$$\Gamma_{\beta,P}(x) = e^{-\beta|x|},$$

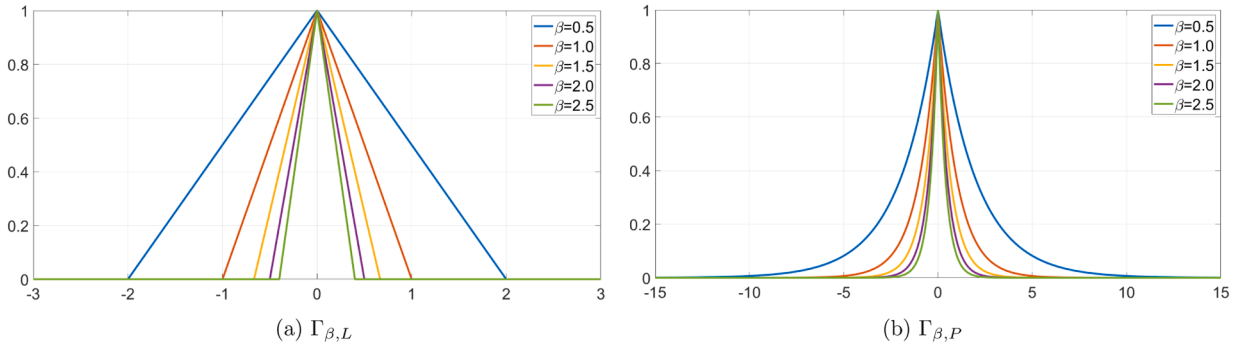


Fig. 1. The S-generations of the fuzzy similarities $\mathcal{E}_{\beta,L}$ and $\mathcal{E}_{\beta,P}$.

is the S-generation of the fuzzy similarity $\mathcal{E}_{\beta,P}$. The S-generations of $\mathcal{E}_{\infty,L}$ and $\mathcal{E}_{\infty,P}$ are the same and given by the function $\Gamma_{\infty} : \mathbb{R} \rightarrow [0, 1]$, where

$$\Gamma_{\infty}(x) = \begin{cases} 1, & x = 0 \\ 0, & \text{otherwise} \end{cases}.$$

We display in Fig. 1 the plots of $\Gamma_{\beta,L}$ and $\Gamma_{\beta,P}$ with different values of β .

For any set $I \subset (0, \infty]$, let us denote by $\mathbb{G}_L^I$ and $\mathbb{G}_P^I$ the sets of S-generations of fuzzy similarities in $\mathbb{E}_L^I$ and $\mathbb{E}_P^I$, respectively.

Definition 5. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be two fuzzy similarities. We say that $\mathcal{E}_1$ is smaller or equal to $\mathcal{E}_2$, denoted by $\mathcal{E}_1 \leq_S \mathcal{E}_2$, if $\mathcal{E}_1(x, y) \leq \mathcal{E}_2(x, y)$, for any $x, y \in \mathbb{R}$.

It is easy to see that $\leq_S$ is a partial order on any set of fuzzy similarities. We say that $\mathcal{E}_1 <_S \mathcal{E}_2$ if $\mathcal{E}_1 \leq_S \mathcal{E}_2$ and $\mathcal{E}_1 \neq \mathcal{E}_2$.

Remark 1. $\mathbb{E}_L^I$ and $\mathbb{E}_P^I$ are linearly ordered sets with respect to $\leq_S$. In particular,

$$\mathcal{E}_{\beta,L} \leq_S \mathcal{E}_{\beta',L} \text{ and } \mathcal{E}_{\beta,P} \leq_S \mathcal{E}_{\beta',P},$$

if and only if $\beta \geq \beta'$.

Definition 6. Let Γ_1 and Γ_2 be two S-generations. We say that Γ_1 is smaller or equal to Γ_2 , denoted by $\Gamma_1 \leq_G \Gamma_2$, if $\Gamma_1(x) \leq \Gamma_2(x)$, for any $x \in \mathbb{R}$.

Lemma 3. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be two fuzzy similarities determined by S-generations Γ_1 and Γ_2 , respectively. Then, $\mathcal{E}_1 \leq_S \mathcal{E}_2$ iff $\Gamma_1 \leq_G \Gamma_2$.

Proof. The proof is straightforward from the connection between a fuzzy similarity and its S-generation. $\square$

Remark 2. $\mathbb{G}_L^I$ and $\mathbb{G}_P^I$ are linearly ordered by $\leq_G$. In particular,

$$\Gamma_{\beta,L} \leq_G \Gamma_{\beta',L} \text{ and } \Gamma_{\beta,P} \leq_G \Gamma_{\beta',P},$$

if and only if $\beta \geq \beta'$.

In the sequel, let us denote by $\mathbb{E}$ a set of fuzzy similarities and $\mathbb{G}$ the set of S-generations of fuzzy similarities in $\mathbb{E}$.

Since a S-generation is a bounded continuous function on $\mathbb{R}$, we can always define the distance between two S-generations, say Γ_1 and Γ_2 , as follows:

$$d_G(\Gamma_1, \Gamma_2) = \sup \{ |\Gamma_1(x) - \Gamma_2(x)| \mid x \in \mathbb{R} \}. \quad (3)$$

Consequently, the set $\mathbb{G}$ with metric d_G defined in (3) forms a metric space. Moreover, $(\mathbb{G}, d_G)$ is a subspace of the metric space $C_{\infty}(\mathbb{R})$.¹

Lemma 4. If I is a closed in $(0, \infty]$, then $(\mathbb{G}_L^I, d_G)$ is a complete metric space.

Proof. Since $C_{\infty}(\mathbb{R})$ is a complete metric space, we only need to prove that $\mathbb{G}_L^I$ is closed in $C_{\infty}(\mathbb{R})$. Indeed, let $(\Gamma_n)_n$, where $\Gamma_n(x) = \max\{1 - \beta_n|x|, 0\}$, $\beta_n \in I$, be a sequence in $\mathbb{G}_L^I$ and $\lim_{n \rightarrow \infty} \Gamma_n = \Gamma$. We show that $\Gamma \in \mathbb{G}_L^I$. Let us consider the following two cases.

First, assume that $\infty \notin I$. It is easy to see that $\Gamma \neq \Gamma_{\infty}$. It follows that there is $0 \neq x_0 \in \mathbb{R}$ such that $\Gamma(x_0) > 0$. Since the convergence of $(\Gamma_n)_n$ in $C_{\infty}(\mathbb{R})$ implies the pointwise convergence of $(\Gamma_n(x))_n$ for any $x \in \mathbb{R}$, there is $N_0 > 0$ such that $\Gamma_n(x_0) > 0$ for any $n > N_0$. It follows that $\Gamma_n(x_0) = 1 - \beta_n|x_0|$ for any $n > N_0$. Consequently, for any $n > N_0$ and $q \in \mathbb{N}^*$, we see that

$$|\Gamma_n(x_0) - \Gamma_{n+q}(x_0)| = |\beta_{n+q} - \beta_n||x_0|$$

¹ The complete metric space of bounded continuous functions on $\mathbb{R}$ with the metric d_{∞} defined by $d_{\infty}(f, g) = \sup \{ |f(x) - g(x)| \mid x \in \mathbb{R} \}$, for any $f, g \in C_{\infty}(\mathbb{R})$.

Since $(\Gamma_n(x_0))_n$ is convergent, it is a Cauchy sequence. It follows that $\lim_{n \rightarrow \infty} |\Gamma_n(x_0) - \Gamma_{n+q}(x_0)| = 0$. Then, we obtain $\lim_{n \rightarrow \infty} |\beta_{n+q} - \beta_n| = 0$. Namely, $(\beta_n)_n$ is a Cauchy sequence in $\mathbb{R}$. Let $\lim_{n \rightarrow \infty} \beta_n = \beta^*$. It follows from the closeness of I that $\beta^* \in I$. We obtain that

$$\lim_{n \rightarrow \infty} \Gamma_n(x) = \lim_{n \rightarrow \infty} \max\{1 - \beta_n|x|, 0\} = \max\{1 - \beta^*|x|, 0\},$$

for any $x \in \mathbb{R}$. As a result, $\Gamma(x) = \max\{1 - \beta^*|x|, 0\}$. Therefore, $\Gamma \in \mathbb{G}_L^I$.

Second, assume that $\infty \in I$. If $\Gamma = \Gamma_\infty$, it is clear that $\Gamma \in \mathbb{G}_L^I$. In the case that $\Gamma \neq \Gamma_\infty$, by the same analysis as above, we obtain that $\Gamma(x) = \max\{1 - \beta^*|x|, 0\}$, with $\beta^* \in I$. Consequently, $\Gamma \in \mathbb{G}_L^I$. $\square$

Lemma 5. *If I is closed in $(0, \infty]$, then $(\mathbb{G}_p^I, d_G)$ is a complete metric space.*

Proof. Similarly to the proof of Lemma 4, we only need to prove that $\mathbb{G}_p^I$ is closed in $C_\infty(\mathbb{R})$. Indeed, let $(\Gamma_n)_n$, where $\Gamma_n(x) = e^{-\beta_n|x|}$, $\beta_n \in I$, be a sequence in $\mathbb{G}_p^I$ and $\lim_{n \rightarrow \infty} \Gamma_n = \Gamma$. We show that $\Gamma \in \mathbb{G}_p^I$. Let us consider the following two cases.

First, assume that $\infty \notin I$. It is easy to see that $\Gamma \neq \Gamma_\infty$. It follows that there is $0 \neq x_0 \in \mathbb{R}$ such that $\Gamma(x_0) > 0$. Since the convergence of $(\Gamma_n)_n$ in $C_\infty(\mathbb{R})$ implies the pointwise convergence of $(\Gamma_n(x))_n$ for any $x \in \mathbb{R}$, $\lim_{n \rightarrow \infty} \Gamma_n(x_0) = \Gamma(x_0) > 0$. It follows that $(\beta_n)_n$ is convergent. Assume that $\lim_{n \rightarrow \infty} \beta_n = \beta^*$. It follows from the closeness of I that $\beta^* \in I$ and

$$\lim_{n \rightarrow \infty} \Gamma_n(x) = \lim_{n \rightarrow \infty} e^{-\beta_n|x|} = e^{-\beta^*|x|},$$

for any $x \in \mathbb{R}$. As a result, $\Gamma(x) = e^{-\beta^*|x|}$. Namely, $\Gamma \in \mathbb{G}_p^I$.

Second, assume that $\infty \in I$. If $\Gamma = \Gamma_\infty$, it is clear that $\Gamma \in \mathbb{G}_p^I$. In the case that $\Gamma \neq \Gamma_\infty$, by the same analysis as above, we obtain that $\Gamma(x) = e^{-\beta^*|x|}$, with $\beta^* \in I$. Consequently, $\Gamma \in \mathbb{G}_p^I$. $\square$

We define the distance between two fuzzy similarities based on their S-generations.

Definition 7. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be two fuzzy similarities determined by S-generations Γ_1 and Γ_2 , respectively. Let us define the distance d_E between $\mathcal{E}_1$ and $\mathcal{E}_2$ as follows:

$$d_E(\mathcal{E}_1, \mathcal{E}_2) = d_G(\Gamma_1, \Gamma_2).$$

Corollary 1. *If I is closed in $(0, \infty]$, then $(\mathbb{E}_L^I, d_E)$ and $(\mathbb{E}_p^I, d_E)$ are complete metric spaces.*

Proof. This is a straightforward consequence of Lemmas 4 and 5. $\square$

3. Calculus on extensional fuzzy numbers

Definition 8. [1] Let $a \in \mathbb{R}$ and $\mathcal{E}$ be a fuzzy similarity in $\mathbb{E}$. Let $a_{\mathcal{E}}$ be defined by $a_{\mathcal{E}}(x) = \mathcal{E}(a, x)$, for any $x \in \mathbb{R}$. We call $a_{\mathcal{E}}$ an *extensional fuzzy number*.

The concept of extensional fuzzy numbers originates from the so-called extensional fuzzy set. Remind that an extensional fuzzy set, say $\mathcal{A}$, of a fuzzy set A on universe U concerning a fuzzy similarity S on $U \times U$ is a fuzzy set on U defined as follows:

$$\mathcal{A}(x) = \max_{u \in U} A(u) \otimes S(u, x), \quad \text{for any } x \in U,$$

where $\otimes$ is the t-norm gathering to the fuzzy similarity S . Under this concept, one can see that extensional fuzzy number $a_{\mathcal{E}}$ is an extensional fuzzy set of the singleton 1_a with respect to fuzzy similarity $\mathcal{E}$, where

$$1_a(x) = \begin{cases} 1, & x = a \\ 0, & \text{otherwise} \end{cases}.$$

Fuzzy numbers, defined this way, are very transparent and follow natural thinking of people. Let us consider an example when someone provides vague information like “about 5” to some quantitative requests. The person receiving this information will immediately think of “number 5” and “a fuzzy region” around this point. In this case, the extensional fuzzy number $5_{\mathcal{E}}$ models exactly what the receiver is visualizing where the similarity $\mathcal{E}$ describes the fuzzy region around 5. Additionally, one can see that a real number is an extensional fuzzy number defined with respect to itself and the crisp equality.

Let $\mathbb{F} = \{a_{\mathcal{E}} \mid a \in \mathbb{R}, \mathcal{E} \in \mathbb{E}\}$. In what follows, we equip to $\mathbb{F}$ a metric.

3.1. Metric space of extensional fuzzy numbers

Definition 9. Let $a_{\mathcal{E}_1}, b_{\mathcal{E}_2}$ be arbitrary in $\mathbb{F}$. The distance between these fuzzy numbers is defined as follows:

$$\mathcal{D}(a_{\mathcal{E}_1}, b_{\mathcal{E}_2}) = |a - b| + d_E(\mathcal{E}_1, \mathcal{E}_2).$$

It is easy to see that $\mathcal{D}$ is a metric on $\mathbb{F}$. Consequently, $(\mathbb{F}, \mathcal{D})$ forms a metric space.

Theorem 1. *If $(\mathbb{E}, d_E)$ is a complete metric space, then $(\mathbb{F}, \mathcal{D})$ is a complete metric space.*

Proof. Let $(a_{\mathcal{E}_n}^n)_n$ be a Cauchy sequence in $\mathbb{F}$ concerning the metric $\mathcal{D}$. It means that for any $\epsilon > 0$, there is $N > 0$ such that $\mathcal{D}(a_{\mathcal{E}_m}^m, a_{\mathcal{E}_n}^n) < \epsilon$, for any $m, n > N$. Since $\mathcal{D}(a_{\mathcal{E}_m}^m, a_{\mathcal{E}_n}^n) = |a^m - a^n| + d_E(\mathcal{E}_m, \mathcal{E}_n)$, we obtain that $|a^m - a^n| < \epsilon$ and $d_E(\mathcal{E}_m, \mathcal{E}_n) < \epsilon$. It follows that $(a^n)_n$ and $(\mathcal{E}_n)_n$ are Cauchy sequences in $\mathbb{R}$ and $\mathbb{E}$, respectively. Since $\mathbb{R}$ and $\mathbb{E}$ are complete metric, there exist $a \in \mathbb{R}$ and $\mathcal{E} \in \mathbb{E}$ such that $\lim_{n \rightarrow \infty} a^n = a$ and $\lim_{n \rightarrow \infty} \mathcal{E}_n = \mathcal{E}$, namely, $\lim_{n \rightarrow \infty} |a^n - a| = 0$ and $\lim_{n \rightarrow \infty} d_E(\mathcal{E}_n, \mathcal{E}) = 0$. It follows that $\lim_{n \rightarrow \infty} \mathcal{D}(a_{\mathcal{E}_n}^n, a_{\mathcal{E}}) = 0$. Therefore, $\lim_{n \rightarrow \infty} a_{\mathcal{E}_n}^n = a_{\mathcal{E}}$. $\square$

For any $I \subset (0, \infty]$, denote $\mathbb{F}_L^I = \{a_{\mathcal{E}} \mid a \in \mathbb{R}, \mathcal{E} \in \mathbb{E}_L^I\}$, and $\mathbb{F}_P^I = \{a_{\mathcal{E}} \mid a \in \mathbb{R}, \mathcal{E} \in \mathbb{E}_P^I\}$. We obtain the following corollary.

Corollary 2. If I is closed in $(0, \infty]$, then $\mathbb{F}_L^I$ and $\mathbb{F}_P^I$ are complete metric spaces.

Proof. The proof is straightforward from Corollary 1 and Theorem 1. $\square$

3.2. Arithmetic operations

Regarding [1], basic arithmetic operations on extensional fuzzy numbers are performed in two phases: 1) calculating the standard operations on their cores which are real numbers and 2) applying the lattice maximum operation to fuzzy similarities. The authors restrict the consideration to extensional fuzzy numbers specified with totally ordered sets of fuzzy similarities. Let us denote by $\mathbb{E}_c$ a set of totally ordered fuzzy similarities and $\mathbb{F}_c$ the set of extensional fuzzy numbers defined with respect to it. We recall the arithmetic of extensional fuzzy numbers in $\mathbb{F}_c$ as follows.

Definition 10. [1] Let $a_{\mathcal{E}_1}, b_{\mathcal{E}_2} \in \mathbb{F}_c$. The addition, subtraction, multiplication and division of $a_{\mathcal{E}_1}$ and $b_{\mathcal{E}_2}$ are defined as follows:

$$\begin{aligned} a_{\mathcal{E}_1} + b_{\mathcal{E}_2} &= (a + b)_{\max\{\mathcal{E}_1, \mathcal{E}_2\}}, \\ a_{\mathcal{E}_1} - b_{\mathcal{E}_2} &= (a - b)_{\max\{\mathcal{E}_1, \mathcal{E}_2\}}, \\ a_{\mathcal{E}_1} \cdot b_{\mathcal{E}_2} &= (ab)_{\max\{\mathcal{E}_1, \mathcal{E}_2\}}, \\ \frac{a_{\mathcal{E}_1}}{b_{\mathcal{E}_2}} &= \left(\frac{a}{b}\right)_{\max\{\mathcal{E}_1, \mathcal{E}_2\}}, \quad b \neq 0. \end{aligned}$$

One can see that calculating of basic operations on extensional fuzzy numbers, as defined in Definition 10, is very simple and can be immediately evaluated by humans. Concerning the addition and multiplication, the neutral and identity elements are real numbers 0 and 1 (extensional fuzzy numbers defined with crisp equality). These operations are associative, commutative and distributive. Namely, for any $a_{\mathcal{E}_1}, b_{\mathcal{E}_2}, c_{\mathcal{E}_3} \in \mathbb{F}_c$, it holds that

$$\begin{aligned} (a_{\mathcal{E}_1} + b_{\mathcal{E}_2}) + c_{\mathcal{E}_3} &= a_{\mathcal{E}_1} + (b_{\mathcal{E}_2} + c_{\mathcal{E}_3}) \\ (a_{\mathcal{E}_1} \cdot b_{\mathcal{E}_2}) \cdot c_{\mathcal{E}_3} &= a_{\mathcal{E}_1} \cdot (b_{\mathcal{E}_2} \cdot c_{\mathcal{E}_3}) \\ a_{\mathcal{E}_1} + b_{\mathcal{E}_2} &= b_{\mathcal{E}_2} + a_{\mathcal{E}_1} \\ a_{\mathcal{E}_1} \cdot b_{\mathcal{E}_2} &= b_{\mathcal{E}_2} \cdot a_{\mathcal{E}_1} \\ (a_{\mathcal{E}_1} + b_{\mathcal{E}_2}) \cdot c_{\mathcal{E}_3} &= a_{\mathcal{E}_1} \cdot c_{\mathcal{E}_3} + b_{\mathcal{E}_2} \cdot c_{\mathcal{E}_3} \end{aligned}$$

Note that $a_{\mathcal{E}} - a_{\mathcal{E}} = 0_{\mathcal{E}}$ and $\frac{b_{\mathcal{E}}}{b_{\mathcal{E}}} = 1_{\mathcal{E}}$, for any $a_{\mathcal{E}}, b_{\mathcal{E}} \in \mathbb{F}_c$, $b \neq 0$. We call $0_{\mathcal{E}}$ and $1_{\mathcal{E}}$, where $\mathcal{E}$ is not the crisp equality, pseudo-neutral and pseudo-identity elements.

Below, let us denote by U an open interval in $\mathbb{R}$.

Definition 11. A fuzzy-valued function $F : U \rightarrow \mathbb{F}_c$ is determined by two functions, $f : U \rightarrow \mathbb{R}$ and $\varphi : U \rightarrow \mathbb{E}_c$, as follows:

$$F(t) = f(t)_{\varphi(t)},$$

for any $t \in U$. We say that F is determined by the pair (f, φ) .

By Definition 11, any real-valued function can be extended to a fuzzy-valued function.

Example 3. Let $\beta(t) = q^{t^2}$, $q > 0$ and let $\varphi(t) = \mathcal{E}_{\beta(t), L}$.

1. Let $F : [-2, 2] \rightarrow \mathbb{F}_L^I$ be defined with respect to the pair (f, φ) , where $f(t) = t^2$. In this case, $F(t) = \mathcal{E}_{\beta(t), L}(t^2, \cdot)$, for any $t \in [-2, 2]$.
2. Let $G : [0, 2] \rightarrow \mathbb{F}_L^I$ be defined with respect to the pair (g, φ) , where $g(t) = e^t$. It is easy to see that $G(t) = \mathcal{E}_{\beta(t), L}(e^t, \cdot)$, for any $t \in [0, 2]$.

In Fig. 2, we display the plot of F and G with $q \in \{0.7, 1.0, 2.0\}$.

Note that one can define functions F and G in Example 3 with respect to fuzzy similarities in $\mathbb{E}_P^I$. In this case, $F(t) = \mathcal{E}_{\beta(t), P}(t^2, \cdot)$, for any $t \in [-2, 2]$, and $G(t) = \mathcal{E}_{\beta(t), P}(e^t, \cdot)$, for any $t \in [0, 2]$. Different effects of these two classes of fuzzy similarities to the plots of F and G are displayed in Fig. 3, where $q = 0.7$.

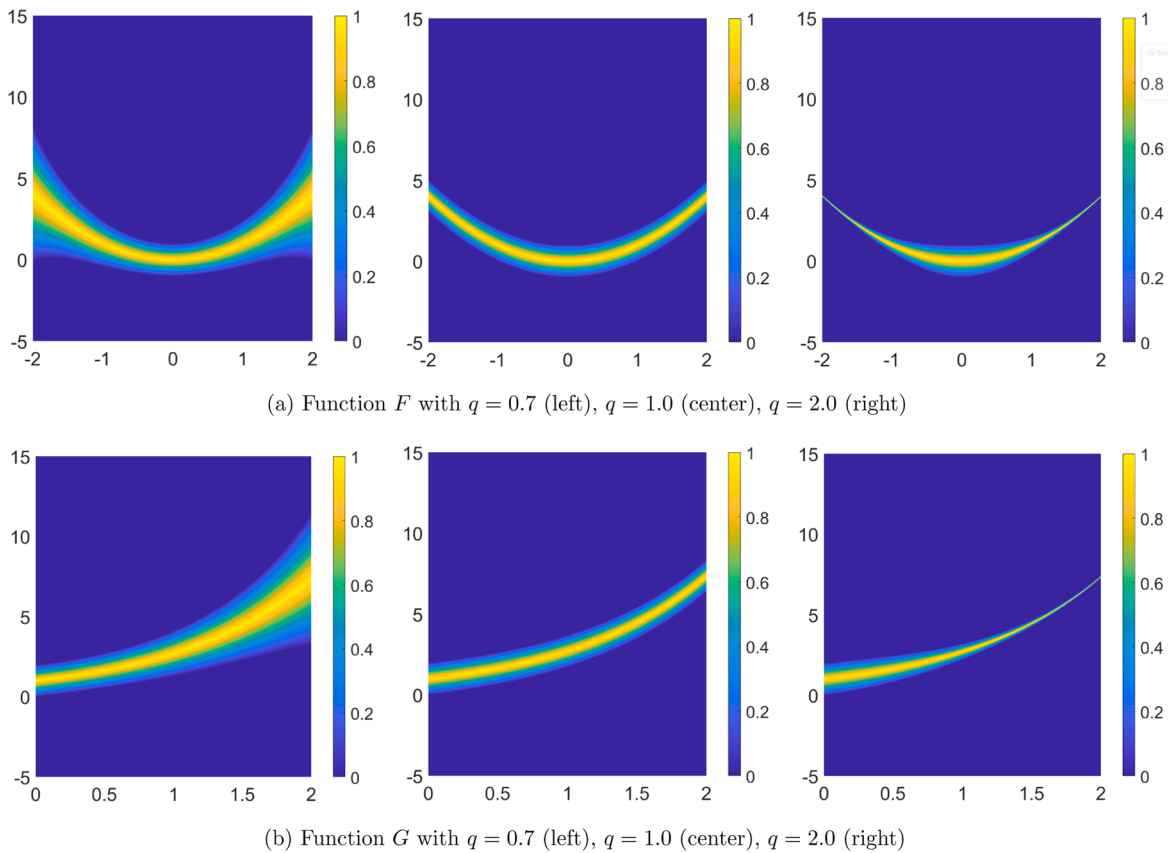


Fig. 2. The plots of functions F and G with different values of q .

3.3. Limit

Definition 12. Let $F : U \rightarrow \mathbb{F}_c$, $t_0 \in U$, and $a_{\mathcal{E}} \in \mathbb{F}_c$. The fuzzy number $a_{\mathcal{E}}$ is said to be the limit of $F(t)$ as t tends to t_0 , denoted by $\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}}$, if for any $\epsilon > 0$, there is $\delta > 0$, such that if $0 < |t - t_0| < \delta$ then $\mathcal{D}(F(t), a_{\mathcal{E}}) < \epsilon$.

From this definition, it is easy to see that

$$\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}} \text{ iff } \lim_{t \rightarrow t_0} \mathcal{D}(F(t), a_{\mathcal{E}}) = 0.$$

Theorem 2. Let $F : U \rightarrow \mathbb{F}_c$ be determined by the pair (f, φ) , $t_0 \in U$, and $a_{\mathcal{E}} \in \mathbb{F}_c$. Then, $\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}}$ iff $\lim_{t \rightarrow t_0} f(t) = a$ and $\lim_{t \rightarrow t_0} \varphi(t) = \mathcal{E}$.

Proof. Assume that $\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}}$. Let $\epsilon > 0$ be arbitrary. Then, there exists $\delta > 0$ such that $0 < |t - t_0| < \delta$ then $\mathcal{D}(F(t), a_{\mathcal{E}}) < \epsilon$. Since $\mathcal{D}(F(t), a_{\mathcal{E}}) = |f(t) - a| + d_E(\varphi(t), \mathcal{E})$, we obtain that $|f(t) - a| < \epsilon$ and $d_E(\varphi(t), \mathcal{E}) < \epsilon$. Therefore, $\lim_{t \rightarrow t_0} f(t) = a$ and $\lim_{t \rightarrow t_0} \varphi(t) = \mathcal{E}$. In the opposite direction, let us assume that $\lim_{t \rightarrow t_0} f(t) = a$ and $\lim_{t \rightarrow t_0} \varphi(t) = \mathcal{E}$. Then, for any $\epsilon > 0$, there exists $\delta > 0$ such that if $0 < |t - t_0| < \delta$ then $|f(t) - a| < \epsilon/2$ and $d_E(\varphi(t), \mathcal{E}) < \epsilon/2$. It follows that $\mathcal{D}(F(t), a_{\mathcal{E}}) = |f(t) - a| + d_E(\varphi(t), \mathcal{E}) < \epsilon$. Hence, we obtain that $\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}}$. $\square$

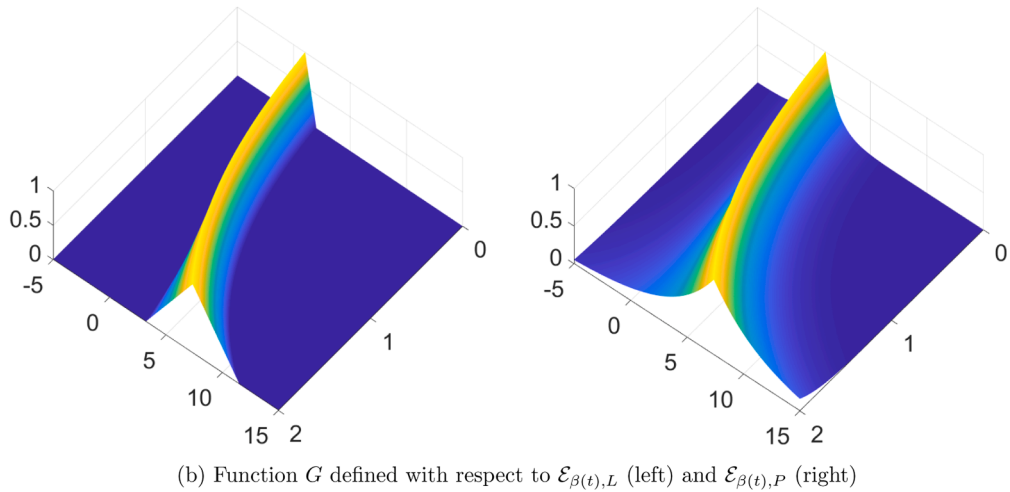
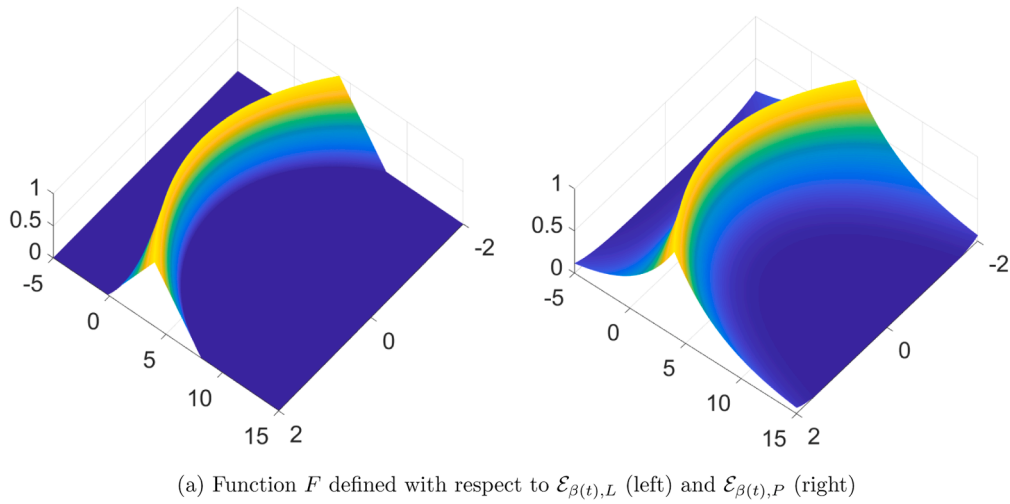
Theorem 3. Let $F : U \rightarrow \mathbb{F}_c$, and let $t_0 \in U$. The limit of $F(t)$ as t tends to t_0 , if it exists, is unique.

Proof. The proof is straightforward from Definition 12 and the properties of metric $\mathcal{D}$. $\square$

Lemma 6. Let $\mathcal{E}_i$, $i = 1, 2, 3$, be fuzzy similarities such that $\mathcal{E}_1 \leq_S \mathcal{E}_2 \leq_S \mathcal{E}_3$. Then, it holds that $d_E(\mathcal{E}_1, \mathcal{E}_2) \leq d_E(\mathcal{E}_1, \mathcal{E}_3)$ and $d_E(\mathcal{E}_1, \mathcal{E}_3) \geq d_E(\mathcal{E}_2, \mathcal{E}_3)$.

Proof. Let Γ_i be the S-generation of $\mathcal{E}_i$, $i = 1, 2, 3$. It follows from Lemma 3 that $\Gamma_1 \leq_G \Gamma_2 \leq_G \Gamma_3$. From (3), we obtain that $d_G(\Gamma_1, \Gamma_2) \leq d_G(\Gamma_1, \Gamma_3)$ and $d_G(\Gamma_1, \Gamma_3) \geq d_G(\Gamma_2, \Gamma_3)$. Therefore, $d_E(\mathcal{E}_1, \mathcal{E}_2) \leq d_E(\mathcal{E}_1, \mathcal{E}_3)$ and $d_E(\mathcal{E}_1, \mathcal{E}_3) \geq d_E(\mathcal{E}_2, \mathcal{E}_3)$. $\square$

Lemma 7. Let $\mathcal{E}_1^0, \mathcal{E}_2^0 \in \mathbb{F}_c$ be such that $\mathcal{E}_1^0 <_S \mathcal{E}_2^0$. For any $\epsilon_i > 0$, $i = 1, 2$, let $B_i = \{\mathcal{E} \in \mathbb{F}_c \mid d_E(\mathcal{E}, \mathcal{E}_i^0) < \epsilon_i\}$. Assume that $\epsilon_1 + \epsilon_2 \leq d_E(\mathcal{E}_1^0, \mathcal{E}_2^0)$. Then, $\mathcal{E}_1 <_S \mathcal{E}_2$, for any $\mathcal{E}_1 \in B_1$ and $\mathcal{E}_2 \in B_2$.

Fig. 3. The plots of functions F and G .

Proof. Assume that there exist $\mathcal{E}_1 \in B_1$, $\mathcal{E}_2 \in B_2$ such that $\mathcal{E}_2 \leq_S \mathcal{E}_1$. Since $\mathbb{E}_c$ is totally ordered, let us consider the following two cases that may occur.

- Case 1: $\mathcal{E}_1^0 \leq_S \mathcal{E}_2$. Then, $\mathcal{E}_1^0 \leq_S \mathcal{E}_2 \leq_S \mathcal{E}_1$. It follows from Lemma 6 that $d_E(\mathcal{E}_2, \mathcal{E}_1^0) \leq d_E(\mathcal{E}_1, \mathcal{E}_1^0) < \epsilon_1$. From $d_E(\mathcal{E}_1^0, \mathcal{E}_2^0) \leq d_E(\mathcal{E}_1^0, \mathcal{E}_2) + d_E(\mathcal{E}_2, \mathcal{E}_2^0)$, we obtain that $d_E(\mathcal{E}_1^0, \mathcal{E}_2^0) < \epsilon_1 + \epsilon_2$. This contradicts the assumption that $d_E(\mathcal{E}_1^0, \mathcal{E}_2^0) \geq \epsilon_1 + \epsilon_2$.
- Case 2: $\mathcal{E}_2 \leq_S \mathcal{E}_1^0$. We see that $\mathcal{E}_2 \leq_S \mathcal{E}_1^0 <_S \mathcal{E}_2^0$. It follows from Lemma 6 that $d_E(\mathcal{E}_1^0, \mathcal{E}_2^0) \leq d_E(\mathcal{E}_2, \mathcal{E}_2^0) < \epsilon_2$. This again contradicts the fact that $d_E(\mathcal{E}_1^0, \mathcal{E}_2^0) \geq \epsilon_1 + \epsilon_2$.

Therefore, $\mathcal{E}_1 <_S \mathcal{E}_2$, for any $\mathcal{E}_1 \in B_1$ and $\mathcal{E}_2 \in B_2$. $\square$

Lemma 8. Let $\varphi, \vartheta : U \rightarrow \mathbb{E}_c$ and $t_0 \in U$. Assume that $\lim_{t \rightarrow t_0} \varphi(t) = \mathcal{E}_1$ and $\lim_{t \rightarrow t_0} \vartheta(t) = \mathcal{E}_2$, with $\mathcal{E}_1, \mathcal{E}_2 \in \mathbb{E}_c$. Then,

$$\lim_{t \rightarrow t_0} \max\{\varphi(t), \vartheta(t)\} = \max\{\mathcal{E}_1, \mathcal{E}_2\}.$$

Proof. We show that $\lim_{t \rightarrow t_0} d_E(\max\{\varphi(t), \vartheta(t)\}, \max\{\mathcal{E}_1, \mathcal{E}_2\}) = 0$. Let $\epsilon > 0$ be arbitrary. In case that $\mathcal{E}_1 = \mathcal{E}_2 = \mathcal{E}$, there is $\delta > 0$ such that if $0 < |t - t_0| < \delta$ then $d_E(\varphi(t), \mathcal{E}) < \epsilon$ and $d_E(\vartheta(t), \mathcal{E}) < \epsilon$. It follows that

$$d_E(\max\{\varphi(t), \vartheta(t)\}, \max\{\mathcal{E}_1, \mathcal{E}_2\}) = d_E(\max\{\varphi(t), \vartheta(t)\}, \mathcal{E}) < \epsilon,$$

for any t such that $0 < |t - t_0| < \delta$. Consequently, we obtain that

$$\lim_{t \rightarrow t_0} d_E(\max\{\varphi(t), \vartheta(t)\}, \max\{\mathcal{E}_1, \mathcal{E}_2\}) = 0.$$

In case that $\mathcal{E}_1 <_S \mathcal{E}_2$, let $d = d_E(\mathcal{E}_1, \mathcal{E}_2)$. Then, there is $\delta > 0$ such that if $0 < |t - t_0| < \delta$ then $d_E(\varphi(t), \mathcal{E}_1) < \min\{\epsilon, d/2\}$ and $d_E(\vartheta(t), \mathcal{E}_2) < \min\{\epsilon, d/2\}$. It follows from Lemma 7 that $\varphi(t) <_S \vartheta(t)$, for any t satisfying that $0 < |t - t_0| < \delta$. Therefore, if $0 < |t - t_0| < \delta$, then

$$d_E(\max\{\varphi(t), \vartheta(t)\}, \max\{\mathcal{E}_1, \mathcal{E}_2\}) = d_E(\vartheta(t), \mathcal{E}_2) < \epsilon.$$

Consequently, $\lim_{t \rightarrow t_0} d_E(\max\{\varphi(t), \vartheta(t)\}, \max\{\mathcal{E}_1, \mathcal{E}_2\}) = 0$. $\square$

Theorem 4. Let $F, G : U \rightarrow \mathbb{F}_c$, $t_0 \in U$, and $a_{\mathcal{E}_1}, b_{\mathcal{E}_2} \in \mathbb{F}_c$. Assume that $\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}_1}$ and $\lim_{t \rightarrow t_0} G(t) = b_{\mathcal{E}_2}$. Then,

- i) $\lim_{t \rightarrow t_0} [F(t) + G(t)] = a_{\mathcal{E}_1} + b_{\mathcal{E}_2}$,
- ii) $\lim_{t \rightarrow t_0} [F(t) - G(t)] = a_{\mathcal{E}_1} - b_{\mathcal{E}_2}$,
- iii) $\lim_{t \rightarrow t_0} F(t) \cdot G(t) = a_{\mathcal{E}_1} \cdot b_{\mathcal{E}_2}$,
- iv) $\lim_{t \rightarrow t_0} \frac{F(t)}{G(t)} = \frac{a_{\mathcal{E}_1}}{b_{\mathcal{E}_2}}$, under additional assumptions that $b \neq 0$ and $G(t)$ is not a pseudo-neutral element, for any $t \in U$.

Proof. Assume that F, G are determined by the pairs (f, φ) and (g, ϑ) , respectively. From $\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}_1}$ and $\lim_{t \rightarrow t_0} G(t) = b_{\mathcal{E}_2}$, we obtain that $\lim_{t \rightarrow t_0} f(t) = a$, $\lim_{t \rightarrow t_0} \varphi(t) = \mathcal{E}_1$, $\lim_{t \rightarrow t_0} g(t) = b$, and $\lim_{t \rightarrow t_0} \vartheta(t) = \mathcal{E}_2$. It follows from Lemma 8 that $\lim_{t \rightarrow t_0} d_E(\max\{\varphi(t), \vartheta(t)\}, \max\{\mathcal{E}_1, \mathcal{E}_2\}) = 0$. Denote by $\star$ one of the addition, subtraction, multiplication, and division. We see that

$$\begin{aligned} \mathcal{D}(F(t) \star G(t), a_{\mathcal{E}_1} \star b_{\mathcal{E}_2}) &= \mathcal{D}([f(t) \star g(t)]_{\max\{\varphi(t), \vartheta(t)\}}, (a \star b)_{\max\{\mathcal{E}_1, \mathcal{E}_2\}}) \\ &= |f(t) \star g(t) - (a \star b)| + d_E(\max\{\varphi(t), \vartheta(t)\}, \max\{\mathcal{E}_1, \mathcal{E}_2\}). \end{aligned}$$

By the properties of the limit of real-valued functions,

$$\lim_{t \rightarrow t_0} |f(t) \star g(t) - (a \star b)| = 0.$$

Hence, we obtain that $\lim_{t \rightarrow t_0} \mathcal{D}(F(t) \star G(t), a_{\mathcal{E}_1} \star b_{\mathcal{E}_2}) = 0$. Namely, $\lim_{t \rightarrow t_0} [F(t) \star G(t)] = a_{\mathcal{E}_1} \star b_{\mathcal{E}_2}$. $\square$

Definition 13. Let $F : U \rightarrow \mathbb{F}_c$, $t_0 \in U$, and $a_{\mathcal{E}} \in \mathbb{F}_c$. The fuzzy number $a_{\mathcal{E}}$ is said to be the left-limit (or right-limit) of $F(t)$ as t tends to t_0 , denoted by $\lim_{t \rightarrow t_0^-} F(t) = a_{\mathcal{E}}$ (or $\lim_{t \rightarrow t_0^+} F(t) = a_{\mathcal{E}}$), if for any $\epsilon > 0$, there is $\delta > 0$, such that if $0 < t_0 - t < \delta$ (or $0 < t - t_0 < \delta$) then $\mathcal{D}(F(t), a_{\mathcal{E}}) < \epsilon$.

As a straightforward consequence of Definitions 12 and 13, we obtain the following theorem.

Theorem 5. Let $F : U \rightarrow \mathbb{F}_c$, $t_0 \in U$. Then,

$$\lim_{t \rightarrow t_0} F(t) = a_{\mathcal{E}} \text{ iff } \lim_{t \rightarrow t_0^-} F(t) = \lim_{t \rightarrow t_0^+} F(t) = a_{\mathcal{E}}.$$

Definition 14. A function $F : U \rightarrow \mathbb{F}_c$ is said to be continuous at $t_0 \in U$, if $\lim_{t \rightarrow t_0} F(t) = F(t_0)$.

Theorem 6. Let $F : U \rightarrow \mathbb{F}_c$ be defined by the pair (f, φ) and $t_0 \in U$. Then, F is continuous at t_0 iff f and φ are continuous at t_0 .

Proof. The proof is immediately obtained from the fact that

$$\lim_{t \rightarrow t_0} F(t) = F(t_0) \text{ iff } \lim_{t \rightarrow t_0} f(t) = f(t_0) \text{ and } \lim_{t \rightarrow t_0} \varphi(t) = \varphi(t_0).$$

$\square$

Example 4. Let $I \subset (0, \infty)$, and $\beta : U \rightarrow I$ be a continuous function on U . Let $f : U \rightarrow \mathbb{R}$ be continuous on U and $\varphi : U \rightarrow \mathbb{E}_L^I$ be defined by $\varphi(t) = \mathcal{E}_{\beta(t), L}$. Then, the fuzzy-valued function $F : U \rightarrow \mathbb{F}_L^I$, defined by the pair (f, φ) , is continuous on U . Indeed, consider function $g : (0, \infty) \rightarrow \mathbb{E}_L^I$, defined by $g(s) = \mathcal{E}_{s, L}$, we see that, for any $s, r \in (0, \infty)$,

$$\begin{aligned} d_E(g(s), g(r)) &= d_E(\mathcal{E}_{s, L}, \mathcal{E}_{r, L}) = d_G(\Gamma_{s, L}, \Gamma_{r, L}) \\ &= \sup \{ |\Gamma_{s, L}(x) - \Gamma_{r, L}(x)| \mid x \in \mathbb{R} \} \\ &= \frac{|s - r|}{\max\{s, r\}}. \end{aligned}$$

It follows that g is continuous on $(0, \infty)$. Since $\beta : U \rightarrow I \subset (0, \infty)$ is continuous on U and $\varphi = g \circ \beta$, φ is continuous on U . It follows from Theorem 6, F is continuous on U .

Example 5. Let I, β and f be defined under the same assumptions as in Example 4. Let $\varphi : U \rightarrow \mathbb{E}_P^I$ be defined by $\varphi(t) = \mathcal{E}_{\beta(t), P}$. Then, the fuzzy-valued function $F : U \rightarrow \mathbb{F}_P^I$, defined by the pair (f, φ) , is continuous on U . Indeed, consider function $g : (0, \infty) \rightarrow \mathbb{E}_P^I$, defined by $g(s) = \mathcal{E}_{s, P}$, we see that

$$d_E(g(s), g(r)) = d_E(\mathcal{E}_{s, P}, \mathcal{E}_{r, P}) = d_G(\Gamma_{s, P}, \Gamma_{r, P})$$

$$= \sup \{ |\Gamma_{s,P}(x) - \Gamma_{r,P}(x)| \mid x \in \mathbb{R} \}$$

$$= \left| e^{-\frac{s \ln \frac{s}{r}}{s-r}} - e^{-\frac{r \ln \frac{s}{r}}{s-r}} \right|,$$

for any $s, r \in (0, \infty)$. Since,

$$\lim_{s \rightarrow r} \frac{s \ln \frac{s}{r}}{s-r} = \lim_{s \rightarrow r} \frac{r \ln \frac{s}{r}}{s-r} = 1,$$

we obtain that $\lim_{s \rightarrow r} d_E(g(s), g(r)) = 0$. Consequently, g is continuous on $(0, \infty)$. By the same analysis as in [Example 4](#), we obtain that φ is continuous on U , and further, F is continuous on U as well.

Definition 15. A function $F : U \rightarrow \mathbb{F}_c$ is said to be left-continuous (or right-continuous) at $t_0 \in U$, if $\lim_{t \rightarrow t_0^-} F(t) = F(t_0)$ (or $\lim_{t \rightarrow t_0^+} F(t) = F(t_0)$).

Definition 16. Let $F : U \rightarrow \mathbb{F}_c$ and $[\nu, \tau] \subset U$. F is said to be continuous on U if it is continuous at every point in U . Additionally, the continuity of F on $[\nu, \tau]$ means that F is continuous on (ν, τ) , right-continuous at ν , and left-continuous at τ .

The following theorem is a straightforward consequence of [Theorem 5](#).

Theorem 7. A function $F : U \rightarrow \mathbb{F}_c$ is continuous at $t_0 \in U$ if and only if it is left-continuous and right-continuous at t_0 .

3.4. Derivative

Definition 17. Let $F : U \rightarrow \mathbb{F}_c$ and $t \in U$. The derivative of F at t , if it exist, is an extensional fuzzy number in $\mathbb{F}_c$ that is denoted by $F'(t)$ and defined by

$$F'(t) = \lim_{h \rightarrow 0} \frac{F(t+h) - F(t)}{h}$$

Assume that F is defined by the pair (f, φ) , then

$$F'(t) = \lim_{h \rightarrow 0} \left[\frac{f(t+h) - f(t)}{h} \right]_{\max\{\varphi(t+h), \varphi(t)\}}.$$

Theorem 8. Let $F : U \rightarrow \mathbb{F}_c$ be defined by the pair (f, φ) and $t \in U$. The derivative of F at t exists iff f is differentiable at t and $\lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t)\}$ exists. Furthermore, if F has a derivative at t , then

$$F'(t) = [f'(t)]_{\mathcal{E}_t},$$

where $\mathcal{E}_t = \lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t)\}$.

Proof. From the fact that

$$\frac{F(t+h) - F(t)}{h} = \left[\frac{f(t+h) - f(t)}{h} \right]_{\max\{\varphi(t+h), \varphi(t)\}},$$

the proof is immediately followed from [Theorem 2](#). $\square$

Corollary 3. Let $F : U \rightarrow \mathbb{F}_c$ be defined by the pair (f, φ) and $t \in U$. If f is differentiable at t and φ is continuous at t , then F has a derivative at t and $F'(t) = [f'(t)]_{\varphi(t)}$.

Example 6. Let $F : U \rightarrow \mathbb{F}_c$ be defined by $F(t) = c_{\mathcal{E}}$, for any $t \in U$, where $c \in \mathbb{R}$, and $\mathcal{E} \in \mathbb{F}_c$. Then, $F'(t) = 0_{\mathcal{E}}$, for any $t \in \mathbb{R}$.

Example 7. Let $\beta(t) = q^{t^2}$, $q > 0$.

1. Let F be a fuzzy-valued function determined by the pair (f, φ) with $f(t) = t^2$ and $\varphi(t) = \mathcal{E}_{\beta(t), L}$ with $q = 0.7$.
2. Let G be a fuzzy-valued function determined by the pair (g, ϑ) with $g(t) = e^{\frac{t^2}{2}}$ and $\vartheta(t) = \mathcal{E}_{\beta(t), P}$ with $q = 0.5$.

It follows from [Examples 4](#) and [5](#) that φ and ϑ are continuous on $\mathbb{R}$. Moreover, f and g are differentiable on $\mathbb{R}$. From [Corollary 3](#), we obtain that F and G have a derivative at any point $t \in \mathbb{R}$. In particular,

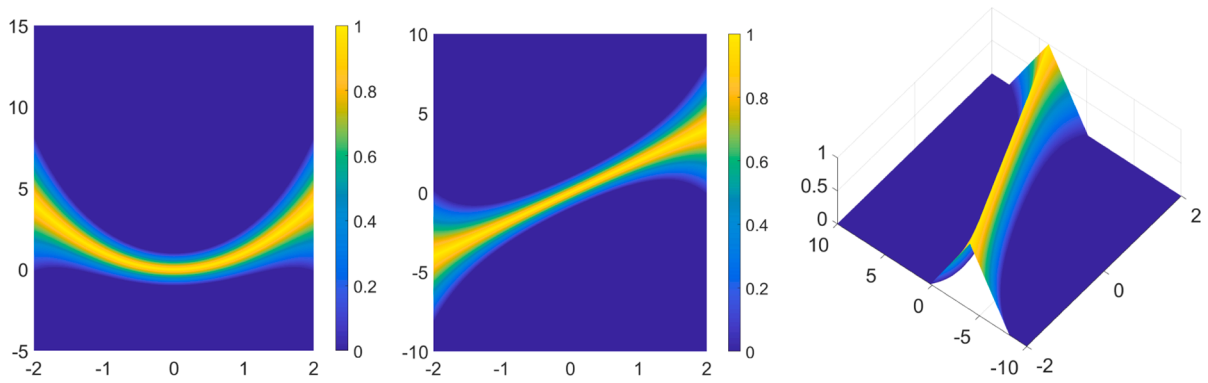
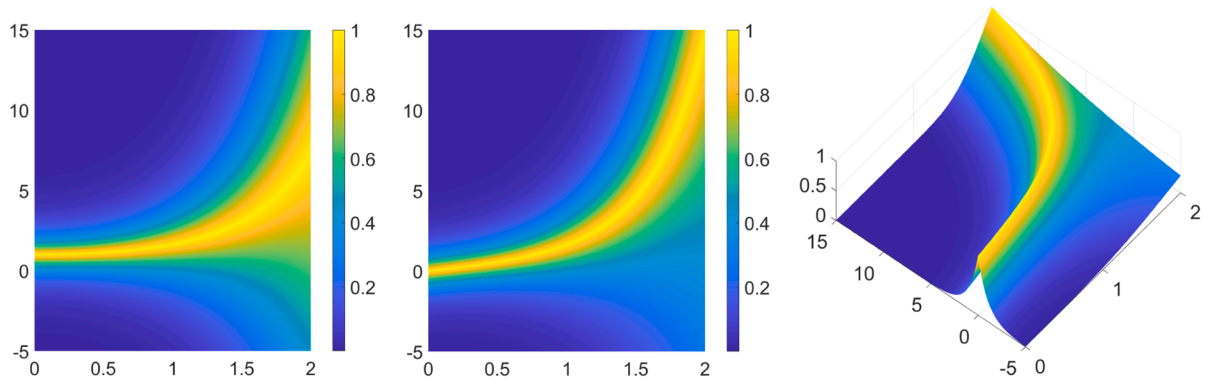
$$F'(t) = [2t]_{\varphi(t)} = \mathcal{E}_{\beta(t), L}(2t, \cdot),$$

and

$$G'(t) = \left[te^{\frac{t^2}{2}} \right]_{\vartheta(t)} = \mathcal{E}_{\beta(t), P}(2e^{2t}, \cdot),$$

for any $t \in \mathbb{R}$. In [Fig. 4](#), we display the plots of $F'(t)$ and $G'(t)$ for $t \in [-2, 2]$ and $t \in [0, 2]$, respectively.

Theorem 9. Let $F : U \rightarrow \mathbb{F}_c$ be defined by the pair (f, φ) and $t \in U$. If F has a derivative at t and φ is continuous at t , then F is continuous at t .

(a) Function F (left) and its derivative in heat plot (center), and in 3D (right)(b) Function G (left) and its derivative in heat plot (center), and in 3D (right)**Fig. 4.** The plots of functions F and G and their derivative.

Proof. Since F has a derivative at t , f is differentiable at t . It follows that f is continuous at t . By φ is continuous at t , we obtain that F is continuous at t . $\square$

Theorem 10. Let $F, G : U \rightarrow \mathbb{F}_c$ be determined by the pairs (f, φ) and (g, ϑ) , respectively. Assume that F and G have derivatives at $t \in U$. Then, $F + G$ has a derivative and

$$(F + G)'(t) = [(f + g)'(t)]_{\mathcal{E}_t},$$

where $\mathcal{E}_t = \lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\}$. Furthermore, if φ and ϑ are continuous at t , then

$$(F + G)'(t) = [(f + g)'(t)]_{\max\{\varphi(t), \vartheta(t)\}}.$$

Proof. We have

$$\begin{aligned} \frac{F(t+h) + G(t+h) - F(t) - G(t)}{h} &= \left[\frac{f(t+h) + g(t+h) - f(t) - g(t)}{h} \right]_{\max\{\varphi(t), \vartheta(t), \varphi(t+h), \vartheta(t+h)\}} \\ &= \left[\frac{f(t+h) - f(t)}{h} + \frac{g(t+h) - g(t)}{h} \right]_{\max\{\varphi(t), \vartheta(t), \varphi(t+h), \vartheta(t+h)\}}. \end{aligned}$$

Since F and G have derivative at t , we obtain that f and g are differentiable at t and $\lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t)\}$ and $\lim_{s \rightarrow t} \max\{\vartheta(s), \vartheta(t)\}$ exist. Moreover, we see that

$$\max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\} = \max\{\max\{\varphi(s), \varphi(t)\}, \max\{\vartheta(s), \vartheta(t)\}\}.$$

It follows from Lemma 8 that

$$\lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\}$$

exists. Consequently,

$$\lim_{h \rightarrow 0} \frac{F(t+h) + G(t+h) - F(t) - G(t)}{h} = [f'(t) + g'(t)]_{\lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\}}$$

$$= [(f + g)'(t)]_{\mathcal{E}_t}.$$

Therefore, $F + G$ has a derivative at t , and further,

$$(F + G)'(t) = [(f + g)'(t)]_{\mathcal{E}_t}.$$

Finally, if φ and ϑ are continuous at t , then it is easy to see that

$$(F + G)'(t) = [(f + g)'(t)]_{\max\{\varphi(t), \vartheta(t)\}}.$$

□

Theorem 11. Let $F, G : U \rightarrow \mathbb{F}_c$ be determined by the pairs (f, φ) and (g, ϑ) , respectively. Assume that F and G have derivatives at $t \in U$. Then, $F \cdot G$ has a derivative and

$$(F \cdot G)'(t) = [(fg)'(t)]_{\mathcal{E}_t},$$

where $\mathcal{E}_t = \lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\}$. Furthermore, if φ and ϑ are continuous at t , then

$$(F \cdot G)'(t) = [(fg)'(t)]_{\max\{\varphi(t), \vartheta(t)\}}.$$

Proof. We have

$$\begin{aligned} \frac{F(t+h)G(t+h) - F(t)G(t)}{h} &= \left[\frac{f(t+h)g(t+h) - f(t)g(t)}{h} \right]_{\max\{\varphi(t), \vartheta(t), \varphi(t+h), \vartheta(t+h)\}} \\ &= \left[\frac{[f(t+h) - f(t)]g(t+h) + f(t)[g(t+h) - g(t)]}{h} \right]_{\max\{\varphi(t), \vartheta(t), \varphi(t+h), \vartheta(t+h)\}}. \end{aligned}$$

Since F and G have derivative at t , f and g are differentiable at t and $\lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t)\}$ and $\lim_{s \rightarrow t} \max\{\vartheta(s), \vartheta(t)\}$ exist. Moreover, we have

$$\max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\} = \max\{\max\{\varphi(s), \varphi(t)\}, \max\{\vartheta(s), \vartheta(t)\}\}.$$

It follows from Lemma 8 that

$$\lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\}$$

exists. We obtain that

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{F(t+h)G(t+h) - F(t)G(t)}{h} &= [f'(t)g(t) + f(t)g'(t)]_{\lim_{s \rightarrow t} \max\{\varphi(s), \varphi(t), \vartheta(s), \vartheta(t)\}} \\ &= [(fg)'(t)]_{\mathcal{E}_t}. \end{aligned}$$

Therefore, $F \cdot G$ has a derivative at t , and further,

$$(F \cdot G)'(t) = [(fg)'(t)]_{\mathcal{E}_t}.$$

Moreover, if φ and ϑ are continuous at t , then

$$(F \cdot G)'(t) = [(fg)'(t)]_{\max\{\varphi(t), \vartheta(t)\}}.$$

□

Theorem 12. Let $F : U \rightarrow \mathbb{F}_c$ be determined by the pair (f, φ) , and let $u : D \rightarrow U$ be a real-valued function. If u is differentiable at $t \in D$ and F has a derivative at $u(t)$, then $F \circ u$ has a derivative at t and

$$(F \circ u)'(t) = u'(t) \cdot F'(u(t))$$

Proof. We see that $F \circ u$ is a fuzzy-valued function defined on D and determined by the pair $(f \circ u, \varphi \circ u)$. Since F has a derivative at $u(t)$, it follows from Theorem 8 that f is differentiable at $u(t)$ and $\lim_{s \rightarrow u(t)} \max\{\varphi(s), \varphi(u(t))\}$ exists. Furthermore, we have that

$$F'(u(t)) = [f'(u(t))]_{\mathcal{E}_{u(t)}},$$

where $\mathcal{E}_{u(t)} = \lim_{\ell \rightarrow u(t)} \max\{\varphi(\ell), \varphi(u(t))\}$. Apply the chain rule with respect to two functions f and u , we obtain that $f \circ u$ is differentiable at t and

$$(f \circ u)'(t) = u'(t) \cdot f'(u(t)). \quad (4)$$

Moreover, we have that

$$\lim_{s \rightarrow t} \max\{(\varphi \circ u)(s), (\varphi \circ u)(t)\} = \lim_{s \rightarrow t} \max\{\varphi(u(s)), \varphi(u(t))\}.$$

By the differentiability of u at t , we obtain that u is continuous at t . It follows that $\lim_{s \rightarrow t} u(s) = u(t)$. Using the fact that $\mathcal{E}_{u(t)} = \lim_{\mathcal{E} \rightarrow u(t)} \max\{\varphi(\mathcal{E}), \varphi(u(t))\}$, we obtain that

$$\lim_{\mathcal{E} \rightarrow u(t)} \max\{(\varphi \circ u)(s), (\varphi \circ u)(t)\} = \mathcal{E}_{u(t)}. \quad (5)$$

It follows from (4), (5) and Theorem 8 that $F \circ u$ has a derivative at t and

$$(F \circ u)'(t) = u'(t) \cdot F'(u(t)).$$

□

Theorem 13. Let $F : [a, b] \rightarrow \mathbb{F}_c$ be defined by the pair (f, φ) . If F is continuous on $[a, b]$ and has derivative at any $t \in (a, b)$, then there is $c \in (a, b)$ such that

$$F'(c) - \frac{F(b) - F(a)}{b - a} = 0_{\mathcal{E}},$$

where $\mathcal{E} = \max\{\varphi(a), \varphi(b), \varphi(c)\}$.

Proof. Since F is continuous on $[a, b]$, it follows from Theorem 6 that f and φ are continuous on $[a, b]$. Moreover, as F has derivative at any $t \in (a, b)$, it follows from Theorem 8 that f is differentiable on (a, b) and further $F'(t) = (f'(t))_{\varphi(t)}$. It follows from the mean value theorem for real-valued functions that there is $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Since $F'(c) = (f'(c))_{\varphi(c)}$, we obtain that

$$F'(c) - \frac{F(b) - F(a)}{b - a} = 0_{\mathcal{E}},$$

where $\mathcal{E} = \max\{\varphi(a), \varphi(b), \varphi(c)\}$. □

Definition 18. Let $F : U \rightarrow \mathbb{F}_c$ and $t \in U$. The left-derivative (or right-derivative) of F at t , if it exists, is an extensional fuzzy number, denoted by $F'_-(t)$ (or $F'_+(t)$) and defined as follows:

$$F'_-(t) = \lim_{h \rightarrow 0^-} \frac{F(t+h) - F(t)}{h},$$

$$F'_+(t) = \lim_{h \rightarrow 0^+} \frac{F(t+h) - F(t)}{h}.$$

The following theorem is a straightforward consequence of Theorem 5.

Theorem 14. A function $F : U \rightarrow \mathbb{F}_c$ has a derivative $F'(t)$ at $t \in U$ if and only if its left-derivative and right-derivative exist and $F'(t) = F'_-(t) = F'_+(t)$.

3.5. Integral

Definition 19. Let $F : [\nu, \tau] \rightarrow \mathbb{F}_c$ and $n \in \mathbb{N}^*$. We divide $[\nu, \tau]$ into n sub-intervals of equal length $\delta_n = \frac{\tau - \nu}{n}$ by points $t_{n,i} = \nu + i\delta_n$, $i = 0, 1, \dots, n$. For any $1 \leq i \leq n$, let $t_{n,i}^* \in [t_{n,i-1}, t_{n,i}]$ be arbitrary. Let

$$S_n = \sum_{i=1}^n F(t_{n,i}^*)\delta_n.$$

F is said to be integrable on $[\nu, \tau]$, if $\lim_{n \rightarrow \infty} S_n$ exists independently on the selection of $t_{n,i}^*$. Then, let us denote

$$\int_{\nu}^{\tau} F(t)dt = \lim_{n \rightarrow \infty} S_n.$$

Theorem 15. Let $F : [\nu, \tau] \rightarrow \mathbb{F}_c$ be defined with respect to the pair (f, φ) . If f is Riemann integrable on $[\nu, \tau]$ and φ is continuous and achieves a maximum at $t_0 \in [\nu, \tau]$, i.e., $\varphi(t_0) = \max\{\varphi(t) \mid t \in [\nu, \tau]\}$, then F is integrable on $[\nu, \tau]$ and

$$\int_{\nu}^{\tau} F(t)dt = \left[\int_{\nu}^{\tau} f(t)dt \right]_{\varphi(t_0)}.$$

Proof. Let S_n , $n \in \mathbb{N}^*$ be defined as in Definition 19, namely,

$$S_n = \sum_{i=1}^n F(t_{n,i}^*)\delta_n.$$

Since F is defined by the pair (f, φ) , we obtain that

$$S_n = \sum_{i=1}^n \left[f(t_{n,i}^*)\delta_n \right]_{\varphi(t_{n,i}^*)} = \left[\sum_{i=1}^n f(t_{n,i}^*)\delta_n \right]_{\max\{\varphi(t_{n,i}^*) \mid i=1, \dots, n\}}.$$

Choose $s_n = t_{n,i}^*$ such that $|s_n - t_0| < \delta_n$. We see that

$$\varphi(s_n) \leq_S \max\{\varphi(t_{n,i}^*) \mid i = 1, \dots, n\} \leq_S \varphi(t_0),$$

and $\lim_{n \rightarrow \infty} s_n = t_0$. Moreover, since φ is continuous at t_0 , we obtain that $\lim_{n \rightarrow \infty} \varphi(s_n) = \varphi(t_0)$. As a straightforward consequence of Lemma 6, we obtain that

$$\lim_{n \rightarrow \infty} \max\{\varphi(t_{n,i}^*) \mid i = 1, \dots, n\} = \varphi(t_0).$$

Moreover, by f is Riemann integrable on $[v, \tau]$, we obtain that

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_{n,i}^*) \delta_n = \int_v^\tau f(t) dt.$$

Therefore,

$$\lim_{n \rightarrow \infty} S_n = \left[\int_v^\tau f(t) dt \right]_{\varphi(t_0)}.$$

Namely, F is integrable on $[v, \tau]$ and

$$\int_v^\tau F(t) dt = \left[\int_v^\tau f(t) dt \right]_{\varphi(t_0)}.$$

□

Example 8. Let $\beta(t) = q t^2$, $q > 0$.

1. Let F be a fuzzy-valued function determined by the pair (f, φ) with $f(t) = t^2$ and $\varphi(t) = \mathcal{E}_{\beta(t), L}$ with $q = 0.7$.
2. Let G be a fuzzy-valued function determined by the pair (g, ϑ) with $g(t) = \ln(t)$ and $\vartheta(t) = \mathcal{E}_{\beta(t), P}$ with $q = 1.5$.

It follows from Example 4 that φ and ϑ are continuous on $\mathbb{R}$. Consider the integrals

$$\int_1^2 F(t) dt, \quad \text{and} \quad \int_1^2 G(t) dt.$$

We have $\int_1^2 f(t) dt = \int_1^2 t^2 dt = \frac{7}{3}$, $\int_1^2 g(t) dt = \int_1^2 \ln t dt = 2 \ln 2 - 1$, $\max\{\varphi(t) \mid t \in [1, 2]\} = \mathcal{E}_{(0.7)^4, L}$, and $\max\{\vartheta(t) \mid t \in [1, 2]\} = \mathcal{E}_{1.5, P}$.

It follows from Theorem 15 that

$$\int_1^2 F(t) dt = \left(\frac{7}{3} \right)_{\mathcal{E}_{(0.7)^4, L}}, \quad \text{and} \quad \int_1^2 G(t) dt = (2 \ln 2 - 1)_{\mathcal{E}_{1.5, P}}.$$

Below, we consider some basic properties of the integral.

Theorem 16. Let $F, G : U \rightarrow \mathbb{F}_c$ be continuous functions, determined by the pairs (f, φ) and (g, ϑ) , respectively. Let $\eta(t) = \max\{\varphi(t), \vartheta(t)\}$, and assume that φ , ϑ and η achieve their maximums on $[v, \tau]$. Then, it holds that

- i) $\int_v^\tau F(t) dt = \int_v^\kappa F(t) dt + \int_\kappa^\tau F(t) dt$, for any $\kappa \in [v, \tau]$,
- ii) $\int_v^\tau c_\mathcal{E} \cdot F(t) dt = c_\mathcal{E} \cdot \int_v^\tau F(t) dt$, for any $c_\mathcal{E} \in \mathbb{F}_c$,
- iii) $\int_v^\tau [F(t) + G(t)] dt + 0_{\max\{\mathcal{E}_1, \mathcal{E}_2\}} = \int_v^\tau F(t) dt + \int_v^\tau G(t) dt$, where $\mathcal{E}_1$ and $\mathcal{E}_2$ are maximums of φ and ϑ on $[v, \tau]$, respectively.

Proof. We have

$$\int_v^\tau f(t) dt = \int_v^\kappa f(t) dt + \int_\kappa^\tau f(t) dt,$$

for any $\kappa \in [v, \tau]$ and

$$\int_v^\tau c \cdot f(t) dt = c \cdot \int_v^\tau f(t) dt,$$

for any $c \in \mathbb{R}$. Therefore, the statements i) and ii) are straightforward consequences of the facts that

$$\max\{\varphi(t) \mid t \in [v, \tau]\} = \max\{\max\{\varphi(t) \mid t \in [v, \kappa]\}, \max\{\varphi(t) \mid t \in [\kappa, \tau]\}\}$$

and $\max\{\max\{\mathcal{E}, \varphi(t)\} \mid t \in [v, \tau]\} = \max\{\mathcal{E}, \max\{\varphi(t) \mid t \in [v, \tau]\}\}$. For the statement iii), let $\mathcal{E}_0 = \max\{\eta(t) \mid t \in [v, \tau]\}$. We see that $\mathcal{E}_0 \leq_S \max\{\mathcal{E}_1, \mathcal{E}_2\}$. It follows that

$$\int_v^\tau [F(t) + G(t)] dt + 0_{\max\{\mathcal{E}_1, \mathcal{E}_2\}} = \left[\int_v^\tau [f(t) + g(t)] dt \right]_{\mathcal{E}_0} + 0_{\max\{\mathcal{E}_1, \mathcal{E}_2\}}$$

$$\begin{aligned}
 &= \left[\int_{\nu}^{\tau} [f(t) + g(t)] dt \right]_{\max\{\mathcal{E}_1, \mathcal{E}_2\}} \\
 &= \left[\int_{\nu}^{\tau} f(t) dt \right]_{\mathcal{E}_1} + \left[\int_{\nu}^{\tau} g(t) dt \right]_{\mathcal{E}_2} \\
 &= \int_{\nu}^{\tau} F(t) dt + \int_{\nu}^{\tau} G(t) dt.
 \end{aligned}$$

□

Definition 20. Let $F : U \rightarrow \mathbb{F}_c$. A function $I_F : U \rightarrow \mathbb{F}_c$ such that $(I_F)'(t) = F(t)$, for any $t \in U$ is called the *antiderivative* of F .

Theorem 17. Let $F : U \rightarrow \mathbb{F}_c$ be defined with respect to the pair (f, φ) . Let $\vartheta : U \rightarrow \mathbb{F}_c$ be a continuous function such that $\vartheta(t) \leq_S \varphi(t)$, for any $t \in U$. If $I_F(t)$ is an antiderivative of $F(t)$, then so is $I_F(t) + C_{\vartheta(t)}$, for any constant $C \in \mathbb{R}$.

Proof. Since, $I_F(t)$ is an antiderivative of $F(t)$, we obtain that

$$[I_F(t) + C_{\vartheta(t)}]' = [I_F(t)]' + [C_{\vartheta(t)}]' = F(t) + 0_{\vartheta(t)} = [f(t)]_{\varphi(t)} + 0_{\vartheta(t)} = [f(t)]_{\max\{\varphi(t), \vartheta(t)\}}.$$

Additionally, we have $\vartheta(t) \leq_S \varphi(t)$, for any $t \in U$. It follows that

$$[I_F(t) + C_{\vartheta(t)}]' = [f(t)]_{\varphi(t)} = F(t).$$

Therefore, $I_F(t) + C_{\vartheta(t)}$ is an antiderivative of $F(t)$. □

Definition 21. Let $F : U \rightarrow \mathbb{F}_c$ be a continuous function, defined with respect to the pair (f, φ) . The indefinite integral of F is denoted by $\int F(t) dt$ and determined as follows:

$$\int F(t) dt = \left[\int f(t) dt \right]_{\varphi(t)}.$$

Theorem 18. Let $F : U \rightarrow \mathbb{F}_c$ be a continuous function, defined with respect to the pair (f, φ) . Then, the function I_F , defined by

$$I_F(t) = \int F(t) dt,$$

is an antiderivative of F .

Proof. It follows from the definition of the indefinite integral of F that

$$I_F(t) = \left[\int f(t) dt \right]_{\varphi(t)}.$$

Since F is continuous on U , φ is continuous too. Therefore, we obtain that

$$[I_F(t)]' = \left[\left(\int f(t) dt \right)' \right]_{\varphi(t)} = [f(t)]_{\varphi(t)} = F(t),$$

for any $t \in U$. Consequently, I_F is an antiderivative of F . □

Theorem 19. Let $F : U \rightarrow \mathbb{F}_c$ be a continuous function, defined with respect to the pair (f, φ) . Let $[\nu, \tau] \subset U$ and assume that φ achieves its maximum on $[\nu, \tau]$. Then, we obtain that

$$\int_{\nu}^{\tau} F(t) dt = I_F(\tau) - I_F(\nu) + 0_{\mathcal{E}},$$

where I_F is defined as in Theorem 18, and $\mathcal{E} = \max\{\varphi(t) | t \in [\nu, \tau]\}$.

Proof. It follows from the continuity of F that f and φ are continuous on $[\nu, \tau]$. Therefore, f is Riemann integrable on $[\nu, \tau]$. We obtain that

$$\int_{\nu}^{\tau} F(t) dt = \left[\int_{\nu}^{\tau} f(t) dt \right]_{\mathcal{E}}.$$

Let $I_f(t) = \int f(t) dt$. We obtain that

$$\int_{\nu}^{\tau} F(t) dt = [I_f(\tau) - I_f(\nu)]_{\mathcal{E}}.$$

Additionally, we see that $I_F(t) = [I_f(t)]_{\varphi(t)}$. It follows that

$$\begin{aligned}
 I_F(\tau) - I_F(\nu) &= [I_f(\tau)]_{\varphi(\tau)} - [I_f(\nu)]_{\varphi(\nu)} \\
 &= [I_f(\tau) - I_f(\nu)]_{\max\{\varphi(\nu), \varphi(\tau)\}}.
 \end{aligned}$$

Since, $\max\{\varphi(\nu), \varphi(\tau)\} \leq_S \mathcal{E}$, we obtain that

$$\int_{\nu}^{\tau} F(t) dt = I_F(\tau) - I_F(\nu) + 0_{\mathcal{E}}.$$

□

Corollary 4. Let $F : U \rightarrow \mathbb{F}_c$ be a continuous function, defined with respect to the pair (f, φ) . Let $[v, \tau] \subset U$ and assume that φ is monotone on $[v, \tau]$. Then, we obtain that

$$\int_v^\tau F(t)dt = I_F(\tau) - I_F(v),$$

where I_F is defined as in [Theorem 18](#).

Proof. Since, φ is monotone on $[v, \tau]$, it achieves a maximum at v or τ . It follows that $\max\{\varphi(v), \varphi(\tau)\} = \max\{\varphi(t) | t \in [v, \tau]\}$. Let $I_f(t) = \int f(t)dt$. Analogously as in the proof of [Theorem 19](#), we obtain that

$$\int_v^\tau F(t)dt = [I_f(\tau) - I_f(v)]_{\max\{\varphi(v), \varphi(\tau)\}}.$$

Moreover,

$$\begin{aligned} I_F(\tau) - I_F(v) &= [I_f(\tau)]_{\varphi(\tau)} - [I_f(v)]_{\varphi(v)} \\ &= [I_f(\tau) - I_f(v)]_{\max\{\varphi(v), \varphi(\tau)\}}. \end{aligned}$$

Consequently,

$$\int_v^\tau F(t)dt = I_F(\tau) - I_F(v).$$

□

Above, we introduced the integral of a fuzzy-valued function through the Riemann sum. Similar to the case of real-valued functions, this approach can be generalized to the Lebesgue type. The remainder of this section is to implement this generalization.

Let $U = [v, \tau]$ and $(U, \mathcal{L}, \mu)$ be the Lebesgue measure space. The central notion of the Lebesgue-type integral of a fuzzy-valued function is the integral of simple functions.

Definition 22. A function $S : U \rightarrow \mathbb{F}_c$ is a simple function if

$$S(t) = \sum_{i=1}^p \Delta[(a_i)_{\mathcal{E}_i}, E_i](t),$$

where $(a_i)_{\mathcal{E}_i} \in \mathbb{F}_c$, $E_i \in \mathcal{L}$, $E_i \cap E_j = \emptyset$ for any $i \neq j$, $\cup_{i=1}^p E_i = U$ and $\Delta[(a_i)_{\mathcal{E}_i}, E_i]$ is a function defined by

$$\Delta[(a_i)_{\mathcal{E}_i}, E_i](t) = \begin{cases} (a_i)_{\mathcal{E}_i}, & t \in E_i \\ 0, & t \notin E_i \end{cases}.$$

The Lebesgue-type integral of the simple function S is defined as follows:

$$\int_U S(t)d\mu = \sum_{i=1}^p (a_i)_{\mathcal{E}_i} \mu(E_i).$$

Remark 3. In [Definition 22](#), if we put $s(t) = \sum_{i=1}^p \Delta[a_i, E_i](t)$ and $\varphi(t) = \mathcal{E}_i$, iff $t \in E_i$, for $i = 1, \dots, p$, then the pair (s, φ) determines the simple function S . Moreover, one can see that

$$\int_U S(t)d\mu = \left[\int_U s(t)d\mu \right]_{\max\{\mathcal{E}_i | i=1, \dots, p\}}.$$

Definition 23. Let $(F_n)_n$ be a sequence of functions $F_n : U \rightarrow \mathbb{F}_c$ determined by the pair (f_n, φ_n) , and let $F : U \rightarrow \mathbb{F}_c$ be determined by (f, φ) , where $\sup\{\varphi(t) | t \in U\}$ exists in $\mathbb{F}_c$. The sequence $(F_n)_n$ is said to be *pointwise consequent from the crisp-side* to F , if the following statements hold:

- i) $\sup\{\varphi_n(t) | t \in U\}$ exists in $\mathbb{F}_c$ and $\sup\{\varphi_n(t) | t \in U\} \leq_S \sup\{\varphi(t) | t \in U\}$,
- ii) $\lim_{n \rightarrow \infty} F_n(t) = F(t)$ for any $t \in U$.

Lemma 9. Let $F : U \rightarrow \mathbb{F}_c$ be determined by the pair (f, φ) , where $\sup\{\varphi(t) | t \in U\}$ exists in $\mathbb{F}_c$. If F is continuous on U , then there is a sequence of fuzzy-valued simple functions defined on U that is pointwise consequent from the crisp-side to F .

Proof. Since F is continuous, f and φ are continuous. It is standard that f is Lebesgue measurable. It follows that there is a sequence of simple functions $s_m : U \rightarrow \mathbb{R}$, with $s_m(t) = \sum_{i=1}^{p_m} \Delta[a_{m,i}, E_{m,i}](t)$, such that $\lim_{m \rightarrow \infty} s_m(t) = f(t)$, for every $t \in U$. For each positive natural number n , divide U into n sub-intervals of equal length $\delta_n = \frac{\tau-v}{n}$ by points $t_{n,j} = v + j\delta_n$, $j = 0, 1, \dots, n$. Let $U_{n,1} = [t_{n,0}, t_{n,1}]$ and $U_{n,j} = (t_{n,j-1}, t_{n,j}]$, for any $1 < j \leq n$. Let

$$\varphi_n(t) = \varphi(t_{n,j}), \text{ iff } t \in U_{n,j}.$$

It is easy to see that $\sup\{\varphi_n(t) \mid t \in U\} \leq_S \sup\{\varphi(t) \mid t \in U\}$. Moreover, for any $t \in U$, we have $d_E(\varphi_n(t), \varphi(t)) = d_E(\varphi(t_{n,j}), \varphi(t))$, where $|t_{n,j} - t| < \delta_n$. Let $\varepsilon > 0$ be arbitrary. By the continuity of φ , there is $\delta > 0$ such that $d_E(\varphi(t'), \varphi(t)) < \varepsilon$, if $|t' - t| < \delta$. Choose N such that $\frac{\varepsilon - \nu}{N} < \delta$. Then, for any $n \geq N$, we obtain that $|t_{n,j} - t| < \delta_n < \delta$. It follows that $d_E(\varphi_n(t), \varphi(t)) < \varepsilon$. Therefore, we obtain that $\lim_{n \rightarrow \infty} \varphi_n(t) = \varphi(t)$. Finally, let

$$S_{m,n}(t) = \sum_{i=1}^{p_m} \sum_{j=1}^n \Delta[(a_{m,i})_{\varphi(t_{n,j})}, E_{m,i} \cap U_{n,j}](t).$$

It is easy to see that this is a fuzzy-valued simple function determined by the pair (s_m, φ_n) . Moreover, as a straightforward consequence of Theorem 2, we obtain that $\lim_{m,n \rightarrow \infty} S_{m,n}(t) = F(t)$. $\square$

Definition 24.

- i) $(\mathbb{E}_c, \leq_S)$ is said to be *left-dense ordering*, provided that for any $\mathcal{E}_1, \mathcal{E}_2 \in \mathbb{E}_c$, $\mathcal{E}_1 <_S \mathcal{E}_2$ and for any $\varepsilon > 0$ there is $\mathcal{E} \in \mathbb{E}_c$ such that $\mathcal{E}_1 <_S \mathcal{E} <_S \mathcal{E}_2$ and $d_S(\mathcal{E}, \mathcal{E}_2) < \varepsilon$.
- ii) $(\mathbb{E}_c, \leq_S)$ is said to be *right-dense ordering*, provided that for any $\mathcal{E}_1, \mathcal{E}_2 \in \mathbb{E}_c$, $\mathcal{E}_1 <_S \mathcal{E}_2$ and for any $\varepsilon > 0$ there is $\mathcal{E} \in \mathbb{E}_c$ such that $\mathcal{E}_1 <_S \mathcal{E} <_S \mathcal{E}_2$ and $d_S(\mathcal{E}, \mathcal{E}_1) < \varepsilon$.
- iii) $(\mathbb{E}_c, \leq_S)$ is said to be *dense ordering* if it is both left-dense ordering and right-dense ordering.

Definition 25. $(\mathbb{E}_c, d_E)$ is said to be *strictly compatible* with the order $\leq_S$, provided that $d_E(\mathcal{E}_2, \mathcal{E}_3) < d_E(\mathcal{E}_1, \mathcal{E}_3)$, for any $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3 \in \mathbb{E}_c$ such that $\mathcal{E}_1 <_S \mathcal{E}_2 \leq_S \mathcal{E}_3$.

Lemma 10. Let $F : U \rightarrow \mathbb{F}_c$ be defined with respect to the pair (f, φ) , where $\sup\{\varphi(t) \mid t \in U\}$ exists in $\mathbb{E}_c$. Assume that the following statements hold

- i) $(\mathbb{E}_c, \leq_S)$ is dense ordering,
- ii) $(\mathbb{E}_c, d_E)$ is strictly compatible with the order $\leq_S$,
- iii) f is Lebesgue measurable on U ,
- iv) $t \mapsto d_E(\varphi(t), \mathcal{E}_0)$, where $\mathcal{E}_0 = \sup\{\varphi(t) \mid t \in U\}$, is Lebesgue measurable on U .

Then, there is a sequence of fuzzy-valued simple functions defined on U that is pointwise consequent from the crisp-side to F .

Proof. Since f is Lebesgue measurable on U , there is a sequence of simple functions

$$s_m(t) = \sum_{i=1}^{p_m} \Delta[a_{m,i}, E_{m,i}](t),$$

such that $\lim_{m \rightarrow \infty} s_m(t) = f(t)$, for every $t \in U$. Moreover, one can see that $0 \leq d_E(\varphi(t), \mathcal{E}_0) \leq 1$, for any $t \in U$. For any positive natural number n , let

$$U_{n,j} = \left\{ t \in U \mid \frac{j-1}{2^n} \leq_S d_E(\varphi(t), \mathcal{E}_0) <_S \frac{j}{2^n} \right\}, \text{ for } j = 1, \dots, 2^n - 1,$$

and let $U_{n,2^n} = \left\{ t \in U \mid \frac{2^n-1}{2^n} \leq d_E(\varphi(t), \mathcal{E}_0) \leq 1 \right\}$. It follows from iv) that $U_{n,j}$ is measurable for any $j = 1, \dots, 2^n$. Let $\varphi_n : U \rightarrow \mathbb{E}$ be defined by

$$\varphi_n(t) = \varphi(t_{n,j}), \text{ if } t \in U_{n,j},$$

where $t_{n,j}$ is fixed in $U_{n,j}$. It is easy to see that

$$\sup\{\varphi_n(t) \mid t \in U\} \leq_S \sup\{\varphi(t) \mid t \in U\}. \quad (6)$$

Moreover, for each $t \in U$,

$$|d_E(\varphi(t), \mathcal{E}_0) - d_E(\varphi_n(t), \mathcal{E}_0)| \leq \frac{1}{2^n}.$$

It follows that

$$\lim_{n \rightarrow \infty} d_E(\varphi_n(t), \mathcal{E}_0) = d_E(\varphi(t), \mathcal{E}_0). \quad (7)$$

Assume that $(\varphi_n(t))_n$ does not converge to $\varphi(t)$. Then, there exists $\varepsilon_0 > 0$ and a subsequence $(\varphi_{n_k}(t))_k$ of $(\varphi_n(t))_n$ such that $d_E(\varphi_{n_k}(t), \varphi(t)) > \varepsilon_0$. Moreover, as $\mathbb{E}_c$ is totally ordered, we can assume that $\varphi_{n_k}(t) <_S \varphi(t) \leq_S \mathcal{E}_0$ or $\varphi(t) <_S \varphi_{n_k}(t) \leq_S \mathcal{E}_0$ for any positive natural number k . In the former case, it follows from the dense ordering of $(\mathbb{E}_c, \leq_S)$ that there is $\mathcal{E} \in \mathbb{E}_c$ such that $\varphi_{n_k}(t) <_S \mathcal{E} <_S \varphi(t) \leq_S \mathcal{E}_0$ for any k . From ii), we obtain that

$$|d_E(\varphi_{n_k}(t), \mathcal{E}_0) - d_E(\varphi(t), \mathcal{E}_0)| > |d_E(\mathcal{E}, \mathcal{E}_0) - d_E(\varphi(t), \mathcal{E}_0)| > 0,$$

for any natural number k , which is in contrast with (7). Similarly, the latter case also leads to a contradiction with (7). Therefore, $(\varphi_n(t))_n$ converges to $\varphi(t)$. Let $S_{m,n} : U \rightarrow \mathbb{F}_c$ be defined by the pair (s_m, φ_n) . Similarly to the proof of Lemma 9, we obtain that $S_{m,n}$ is a fuzzy-valued simple function that is pointwise convergent to F . $\square$

Definition 26. A fuzzy-valued function $F : U \rightarrow \mathbb{F}_c$ determined by the pair (f, φ) is said to be *f-measurable*, if the following statement holds:

- i) $\sup\{\varphi(t) \mid t \in U\}$ exists in $\mathbb{E}_c$,
- ii) There is a sequence of fuzzy-valued simple functions that is pointwise consequent from the crisp-side to F .

Corollary 5. Let $F : U \rightarrow \mathbb{F}_c$ be defined with respect to the pair (f, φ) , where $\mathbb{F}_c$ is $\mathbb{F}_L^I$ or $\mathbb{F}_P^I$. Then F is *f-measurable* if the following statements hold:

- i) $\sup\{\varphi(t) \mid t \in U\}$ exists in $\mathbb{E}_c$,
- ii) f is Lebesgue integrable on U ,
- iii) $t \mapsto d_E(\varphi(t), \mathcal{E}_0)$, where $\mathcal{E}_0 = \sup\{\varphi(t) \mid t \in U\}$, is Lebesgue measurable on U .

Proof. The proof is straightforward using the fact that $(\mathbb{F}_L^I, \leq_S)$ and $(\mathbb{F}_P^I, \leq_S)$ are dense ordering and $(\mathbb{F}_L^I, d_E)$ and $(\mathbb{F}_P^I, d_E)$ are strictly compatible with the order $\leq_S$. $\square$

Definition 27. Let $F : U \rightarrow \mathbb{F}_c$ be a *f-measurable* function, and let $(S_n)_n$ be a sequence of fuzzy-valued simple functions that is pointwise consequent from the crisp-side to F . The Lebesgue-type integral of F on U is defined as follows:

$$\int_U F(t) d\mu = \lim_{n \rightarrow \infty} \int_U S_n(t) d\mu.$$

F is said to be *Lebesgue-type integrable* on U if its Lebesgue-type integral on U is a fuzzy number in $\mathbb{F}_c$.

The following lemma is to show that the Lebesgue-type integral of an Lebesgue-type integrable fuzzy-valued function does not depend on the construction of the sequence of fuzzy-valued simple functions.

Lemma 11. Let $F : U \rightarrow \mathbb{F}_c$ be a Lebesgue-type integrable function, and let $(S_n)_n$ and $(V_n)_n$ be two sequences of fuzzy-valued simple functions that pointwise consequent from the crisp-side to F . Then,

$$\int_U F(t) d\mu = \lim_{n \rightarrow \infty} \int_U S_n(t) d\mu = \lim_{n \rightarrow \infty} \int_U V_n(t) d\mu.$$

Proof. Assume that F is determined by the pair (f, φ) , $S_n(t) = \sum_{i=1}^{p_n} \Delta[(a_{n,i})_{\mathcal{E}_{n,i}}, E_{n,i}](t)$, and $V_n(t) = \sum_{i=1}^{q_n} \Delta[(b_{n,i})_{\mathcal{E}'_{n,i}}, U_{n,i}](t)$. Let $s_n(t) = \sum_{i=1}^{p_n} \Delta[a_{n,i}, E_{n,i}](t)$, $\varphi_n(t) = \sum_{i=1}^{p_n} \Delta[\mathcal{E}_{n,i}, E_{n,i}](t)$, $v_n(t) = \sum_{i=1}^{q_n} \Delta[b_{n,i}, U_{n,i}](t)$, and $v_n(t) = \sum_{i=1}^{q_n} \Delta[\mathcal{E}'_{n,i}, U_{n,i}](t)$. It follows from Remark 3 that S_n and V_n are determined by the pair (s_n, φ_n) and (v_n, v_n) , respectively. Furthermore, we obtain that

$$\int_U S_n(t) d\mu = \left[\int_U s_n(t) d\mu \right]_{\max\{\mathcal{E}_{n,i} \mid i=1, \dots, p_n\}} = \left[\int_U s_n(t) d\mu \right]_{\max\{\varphi_n(t) \mid t \in U\}}$$

and

$$\int_U V_n(t) d\mu = \left[\int_U v_n(t) d\mu \right]_{\max\{\mathcal{E}'_{n,i} \mid i=1, \dots, q_n\}} = \left[\int_U v_n(t) d\mu \right]_{\max\{v_n(t) \mid t \in U\}}.$$

By the pointwise convergence from the crisp-side of $S_n(t)$ and $V_n(t)$ to $F(t)$ and using Theorem 2, we obtain the following statements:

$$\lim_{n \rightarrow \infty} s_n(t) = f(t), \quad \lim_{n \rightarrow \infty} v_n(t) = f(t), \quad (8)$$

$$\lim_{n \rightarrow \infty} \varphi_n(t) = \varphi(t), \quad \lim_{n \rightarrow \infty} v_n(t) = \varphi(t), \quad (9)$$

$$\max\{\varphi_n(t) \mid t \in U\} \leq_S \sup\{\varphi(t) \mid t \in U\}, \quad (10)$$

and

$$\max\{v_n(t) \mid t \in U\} \leq_S \sup\{\varphi(t) \mid t \in U\}. \quad (11)$$

It follows from (8) and the definition of Lebesgue integral of real-valued functions that

$$\lim_{n \rightarrow \infty} \int_U s_n(t) d\mu = \lim_{n \rightarrow \infty} \int_U v_n(t) d\mu = \int_U f(t) d\mu. \quad (12)$$

It follows from (10) that

$$\varphi_n(t) \leq_S \max\{\varphi_n(t) \mid t \in U\} \leq_S \sup\{\varphi(t) \mid t \in U\},$$

for any $t \in U$. As n tends to infinity, we obtain that

$$\varphi(t) \leq_S \lim_{n \rightarrow \infty} \max\{\varphi_n(t) \mid t \in U\} \leq_S \sup\{\varphi(t) \mid t \in U\},$$

for any $t \in U$. It yields that

$$\lim_{n \rightarrow \infty} \max\{\varphi_n(t) \mid t \in U\} = \sup\{\varphi(t) \mid t \in U\}.$$

Similarly, we can show that

$$\lim_{n \rightarrow \infty} \max\{v_n(t) \mid t \in U\} = \sup\{\varphi(t) \mid t \in U\}.$$

Therefore,

$$\lim_{n \rightarrow \infty} \max\{\varphi_n(t) \mid t \in U\} = \lim_{n \rightarrow \infty} \max\{v_n(t) \mid t \in U\} = \sup\{\varphi(t) \mid t \in U\}. \quad (13)$$

From (12) and (13) and Theorem 2, we obtain that

$$\lim_{n \rightarrow \infty} \int_U S_n(t) d\mu = \lim_{n \rightarrow \infty} \int_U V_n(t) d\mu = \left[\int_U f(t) d\mu \right]_{\sup\{\varphi(t) \mid t \in U\}}.$$

□

From the proof of Lemma 11, we can obtain the following corollaries for evaluating of the Lebesgue-type integral of Lebesgue-type integrable functions.

Corollary 6. Let $F : U \rightarrow \mathbb{F}_c$ be a Lebesgue-type integrable function determined with respect to the pair (f, φ) . Then,

$$\int_U F(t) d\mu = \left[\int_U f(t) d\mu \right]_{\sup\{\varphi(t) \mid t \in U\}}.$$

Corollary 7. Let $F : U \rightarrow \mathbb{F}_c$ be determined with respect to the pair (f, φ) , and suppose that the assumptions in Theorem 15 are satisfied. Then,

$$\int_U F(t) d\mu = \left[\int_U f(t) dt \right]_{\varphi(t_0)},$$

where $t_0 \in U$ is the point at which φ obtains its maximum value.

4. Fuzzy initial value problem with the proposed calculus

Let us consider n th order fuzzy differential equation of the following form:

$$\mathcal{X}^{(n)} = F(\mathcal{X}^{(n-1)}, \mathcal{X}^{(n-2)}, \dots, \mathcal{X}, t), \quad t \in U, \quad (14)$$

where $\mathcal{X}$ is an unknown fuzzy-valued function, $\mathcal{X}^{(k)}$, $k = 1, \dots, n$, is the k th order derivative of $\mathcal{X}$ and $F(\mathcal{X}^{(n-1)}, \mathcal{X}^{(n-2)}, \dots, \mathcal{X}, t)$ is an analytic expression of $\mathcal{X}^{(n-1)}, \mathcal{X}^{(n-2)}, \dots, \mathcal{X}$, and t with the four fuzzy arithmetic operations. A function $\mathcal{X} : U \rightarrow \mathbb{F}_c$ is said to be a solution of (14), if it is continuous, has derivatives up to order n on U and together with its derivatives it satisfies (14). Eq. (14) gathering with the following conditions:

$$\mathcal{X}^{(k)}(t_0) = (a_k)_{\mathcal{E}_k}, \quad k = 0, 1, \dots, n, \quad (15)$$

formulating the known (fuzzy) data of a phenomenon at the starting point t_0 , is known as an initial value problem. A solution of (14) that satisfies (15) is called the solution of the initial problem.

4.1. A comparison with existence approaches

Recently, in [15], the authors have devoted a heavy survey with valuable comparisons between most of approaches in investigating fuzzy differential equations including methods that use fuzzy derivative concepts (see [7,16,17] and elsewhere) and those that do not (see [2,18–20], and other places). According to this survey, the fuzzy derivative-based approach faces several difficulties that can not make the theory of fuzzy differential equations move forward unless the used derivative is defined without the relation with the Minkowski addition. In contrast, the studies using the Zadeh's extension principle and fuzzy bunch of real-valued functions with the classical derivative show their efficiency in this direction. A comparison between these methods has been considered in [15] over particular examples.

In what follows, picking a specific initial value problem provided in [15], we demonstrate how the proposed calculus works on it and discuss on the difference between our approach and the approach using fuzzy bunch of real-valued functions, proposed in [18,19].

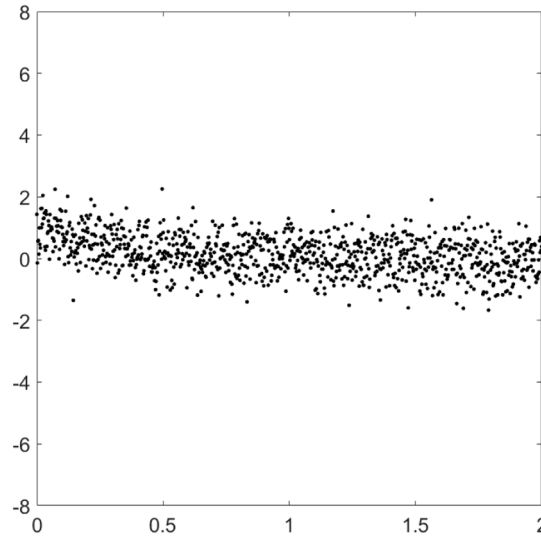


Fig. 5. Observed data.

Consider Example 4.2 in [15], the problem in this example can be translated to the following problem by tools in the proposed calculus. Particularly,

$$\begin{cases} \mathcal{X}'' = 3\mathcal{X} - 2\mathcal{X}' \\ \mathcal{X}(0) = 1_{\mathcal{E}^*} \\ \mathcal{X}'(0) = (-3)_{\mathcal{E}^*} \end{cases}, \quad (16)$$

where the unknown function $\mathcal{X}$ values in $\mathbb{F}_L^I$, for $I = (0, \infty]$, and $\mathcal{E}^* = \mathcal{E}_{1,L}$. It follows from the proposed calculus that a fuzzy-valued function $\mathcal{X}$ determined by the pair (x, φ) , i.e., $\mathcal{X}(t) = [x(t)]_{\varphi(t)}$, is a solution of (16) if and only if

$$\begin{cases} x'' = 3x - 2x' \\ x(0) = 1 \\ x'(0) = -3 \end{cases},$$

and $\varphi(0) = \mathcal{E}^*$. Therefore, it is easy to see that $x(t) = e^{-3t}$. However, let us note that there are infinitely many functions $\varphi : [0, \infty) \rightarrow \mathbb{F}_L^I$ satisfying that $\varphi(0) = \mathcal{E}^*$, e.g., $\varphi(t) = \mathcal{E}_{\beta(t), L}$, where $\beta(t) = q^t$, for any $q > 0$. As a result, (16) has infinitely many solutions under the proposed calculus. Those solutions have the same core $x(t)$ but they differ in the fuzzy similarity relations $\varphi(t)$. Denote by $\mathbf{S}$ the set of solutions of (16), then

$$\mathbf{S} = \left\{ \mathcal{X}(t) = [e^{-3t}]_{\varphi(t)} \mid \varphi : [0, \infty) \rightarrow \mathbb{F}_L^I, \varphi \text{ is continuous, and } \varphi(0) = \mathcal{E}^* \right\}.$$

In [15], using fuzzy bunch of real-valued functions, the authors obtain the solution

$$\mathcal{X}_b(t) = (e^{-3t} - e^t, e^{-3t}, e^t + e^{-3t}), \quad (17)$$

where the right-hand side stands for a triangular fuzzy number.² Moreover, they point out that this solution coincides with the solutions obtained by the generalized Hukuhara derivative approach and by the Zadeh's extension principle (a summary of these methods can be found in [15]). It is easy to see that $\mathcal{X}_b \in \mathbf{S}$, since it is determined by the pair (e^{-3t}, φ) with $\varphi(t) = \mathcal{E}_{e^t, L}$.

For a glance, the existence of infinitely many solutions to Problem (16) may lead to serious difficulties for both theoretical and application research on fuzzy differential equations with the proposed calculus. However, this property is fortunately what we need to provide a promising study direction for using fuzzy differential equations in practice. Below, we describe this contribution based on a practical application of Problem (16).

Consider the situation where someone would like to find a model for a phenomenon. The exact model is unknown, and only observed data of the phenomenon are available, as shown in the Fig. 5. They decide to use Problem (16) to find a fuzzy model

² A triangular fuzzy number $\mathcal{U} = (A, B, C)$, where $A \leq B \leq C$, means that the membership degree of $\mathcal{U}$ is as follows:

$$\mathcal{U}(s) = \begin{cases} \frac{s-A}{B-A}, & A < s < B \\ 1, & s = B \\ \frac{C-s}{C-B}, & B < s < C \\ 0, & \text{otherwise} \end{cases}$$

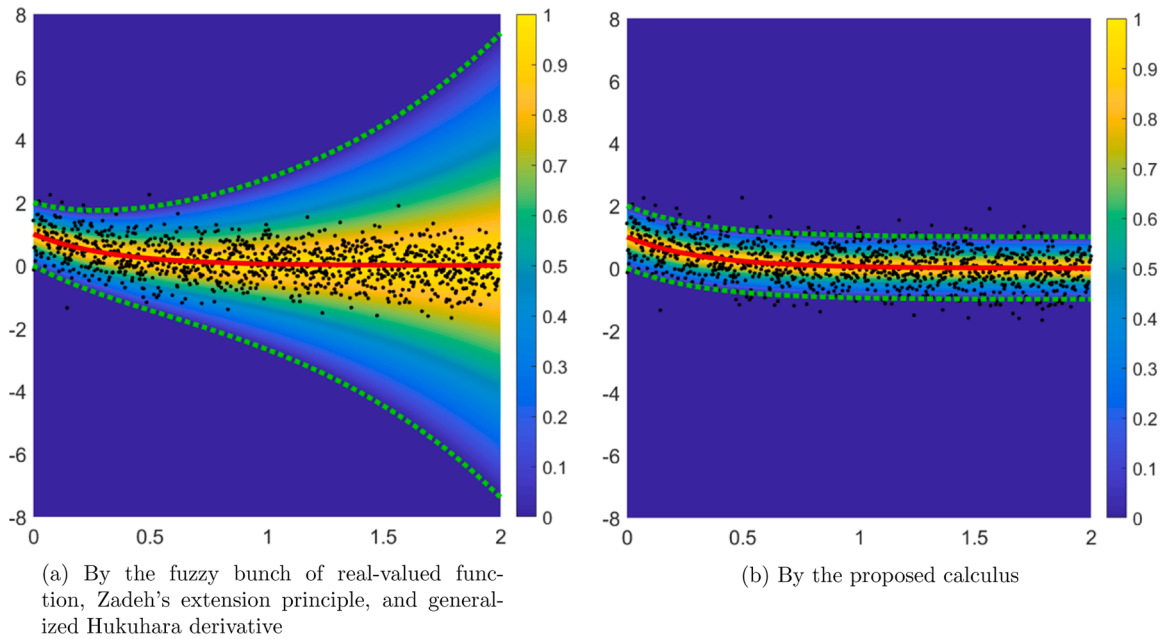


Fig. 6. A comparison on solutions of problem (16) obtained by the proposed calculus and benchmark approaches.

of it. Solving this problem by the bunch of real-valued functions, Zadeh's extension principle or generalized Hukuhara derivative approaches, they get a unique solution as displayed in Fig. 6a. One can see that this solution indicates that the uncertainty of it increases quickly over time (by observing on the dotted green lines). However, when looking at the data, it is quite clear that the uncertainty seems to be unchanged over time (in statistics, they call homoscedastic data [21]). This leads to the fact that the obtained solution is quite far from the observed data. In contrast, formulating and solving Problem (16) with respect to the proposed calculus results in the set S of solutions that differ from each other with respect to the function φ , which describes the dependence of the uncertainty of solutions on time. Since the observation that the uncertainty of the data seems to be unchanged over time, it is reasonable to assume that φ is a constant function. Then, from the initial condition $\varphi(0) = \mathcal{E}^*$, we obtain that $\varphi(t) = \mathcal{E}^*$. Consequently, we obtain a solution in S , depicted in Fig. 6b that fits the observed data better than the solution obtained by the benchmark methods. Note that finding function φ as done relies mostly on descriptive statistics. In situations where the observed data is more complex, the uncertainty changes on time (in statistics, they call heteroscedastic data [21]), statistical tools can be used to assess the degree of variability and uncertainty of the data, e.g., Breusch and Pagan test [22], White test [23]. Then, function φ can be estimated based on regression models. We plan to present this in more details in subsequent studies.

4.2. A practical model with the proposed calculus

In this subsection, we consider an example of using the proposed calculus to establish a mathematical model for the population growth problem.

4.2.1. Theoretical analysis

Knowing that modeling the variation in the number of individuals in a population has never been easy. It depends on various influences such as birth and death rates, immigration and emigration, living environment changes, and many other factors. Let us consider the simplest case where the total number of individuals depends only on birth and death rates. The widely used mathematical model is the following differential equation,

$$x'(t) = \theta x(t), \quad (18)$$

where $x(t)$ and θ are the number of individuals at time t and the growth rate in the population, respectively. This problem has solutions of the following form, $x(t) = Ce^{\theta t}$, where C is a constant. Although solution $x(t)$ gives us an exact number of individuals in the population at a certain time, in practice, this result is only used to estimate the approximate number of individuals in the population because everyone knows that reaching an accurate number of individuals is impossible. In addition, solution $x(t)$ is also uncertain because the growth rate θ cannot be known exactly. A usual description is, for example, "the population has about a individuals at time t " or "the growth rate is approximately θ ". Instead of building an "exact model" as in (18), one can take uncertain information into the model and extend it to a fuzzy version. The following equation is a fuzzy model of (18) based on the calculus of extensional fuzzy numbers,

$$\mathcal{X}'(t) = \theta_{\mathcal{E}} \mathcal{X}(t), \quad (19)$$

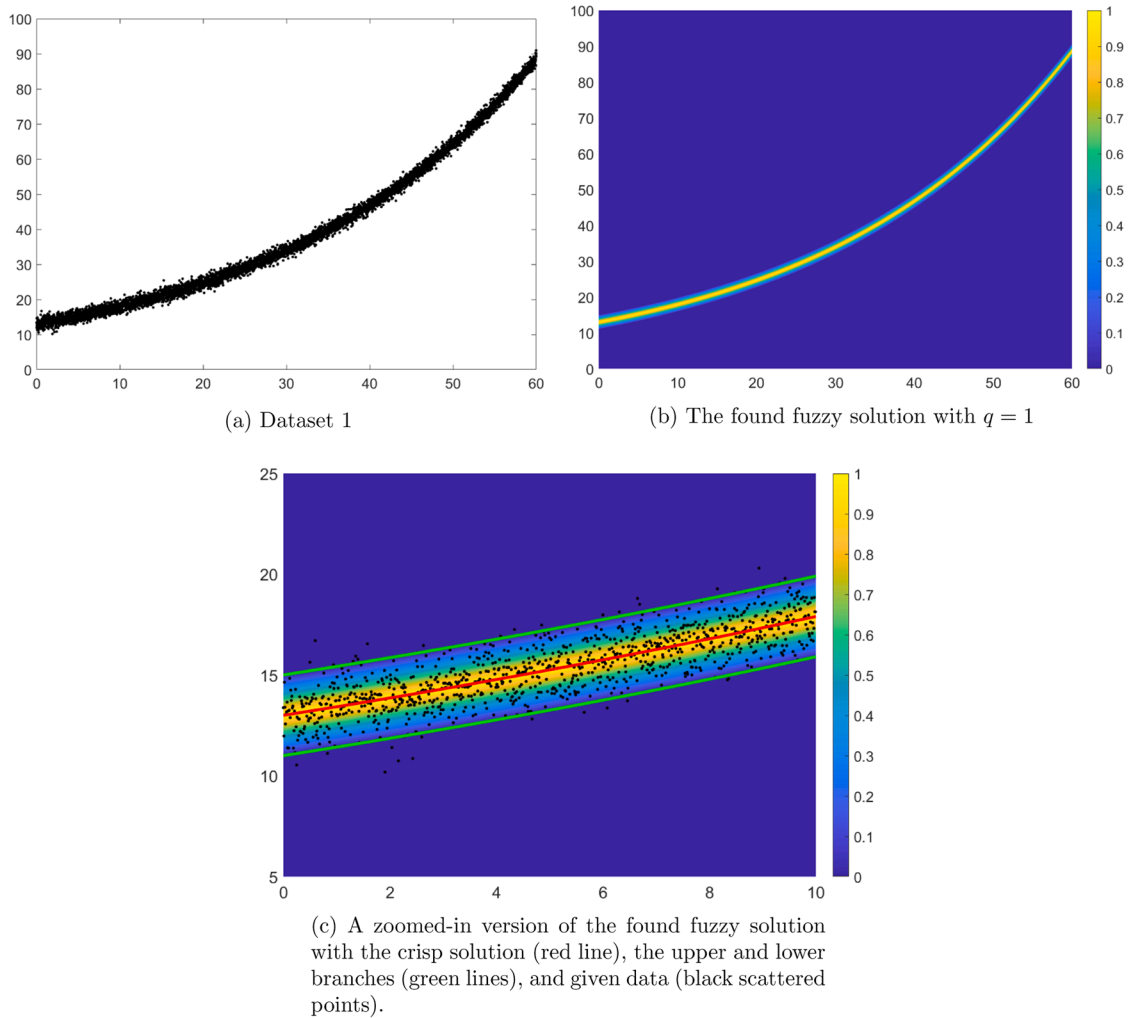


Fig. 7. Simulation results for Dataset 1.

where $\theta_{\mathcal{E}}$ is an extensional fuzzy number, $\mathcal{X}(t)$ is a fuzzy valued function defined on a real interval. Assume that $\mathcal{X}(t) = [x(t)]_{\varphi(t)}$ is a solution of the problem, i.e.,

$$[x'(t)]_{\varphi(t)} = \theta_{\mathcal{E}}[x(t)]_{\varphi(t)}.$$

It follows that $x(t)$ is a solution of (18) and $\varphi(t) \geq_S \mathcal{E}$, for any time t . This means that the “uncertainty level” of the solution $\mathcal{X}(t)$ cannot be lower than the uncertainty of the growth rate (the only known uncertain component of the problem).

Now, let us discuss how to deal with an initial condition to Eq. (19). For Eq. (18), the initial condition is usually expressed by $x(0) = x_0$, where x_0 is a real number. It means that the number of individuals in the population at the initial time is x_0 . The solution of (18) under this condition is

$$x(t) = x_0 e^{\theta t}.$$

As mentioned, it is not easy to obtain an exact value as x_0 . A popular statement of the initial condition is “the number of individuals at the initial time is approximately x_0 ”. Consequently, the initial condition of (19) should be expressed as,

$$\mathcal{X}(0) = (x_0)_{\mathcal{E}^*}, \quad (20)$$

where $\mathcal{E}^*$ is a fuzzy similarity describing the uncertainty of x_0 . Since the number of individuals depends on the measurement or counting unit while the growth rate is not, we can always assume that $\mathcal{E}^* \geq_S \mathcal{E}$ ³. Consequently, the set of solutions of (19) under the

³ When modeling fuzzy information as “about 10 thousand”, where “thousand” is a counting unit, we can convert it to the number 10,000 and then model the information “about10000” instead of the original. Modeling “about 10 thousand” and “about10000” by extensional fuzzy numbers

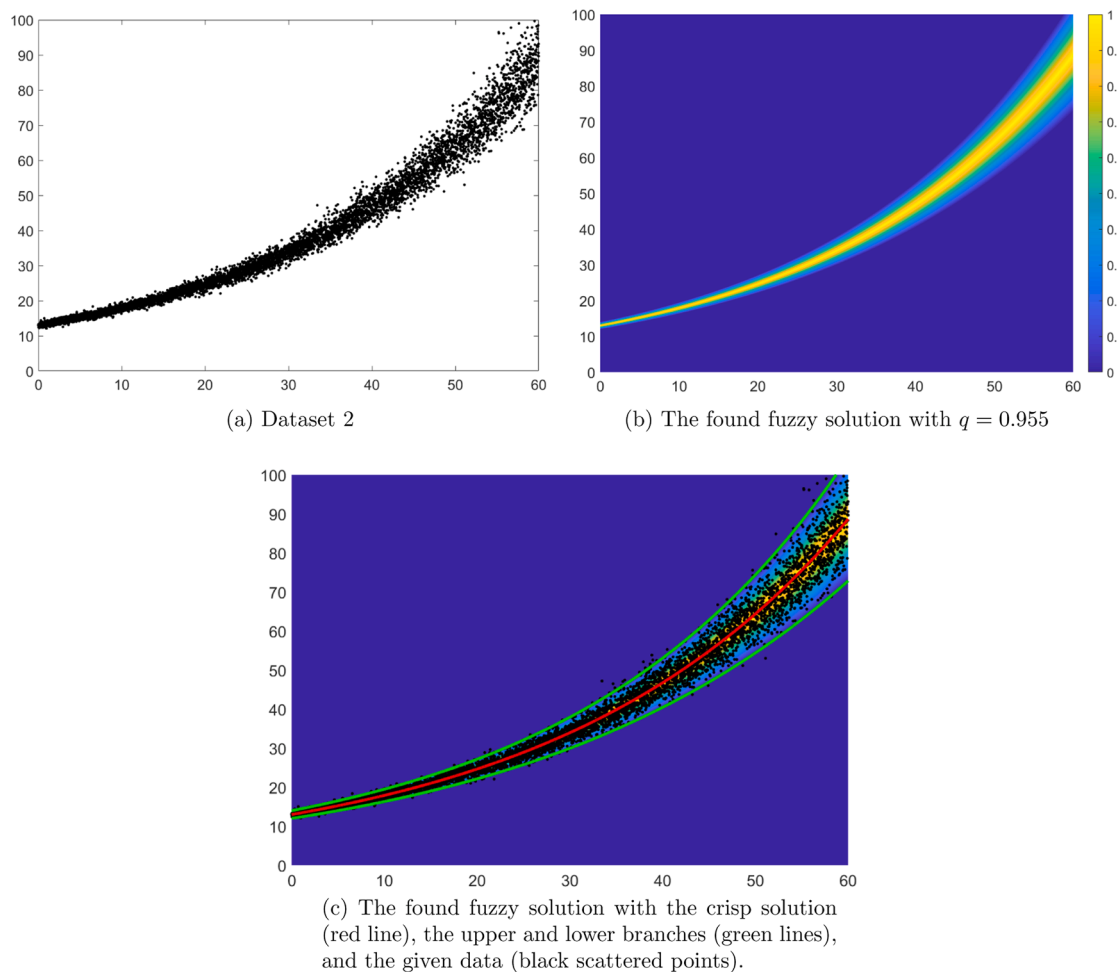
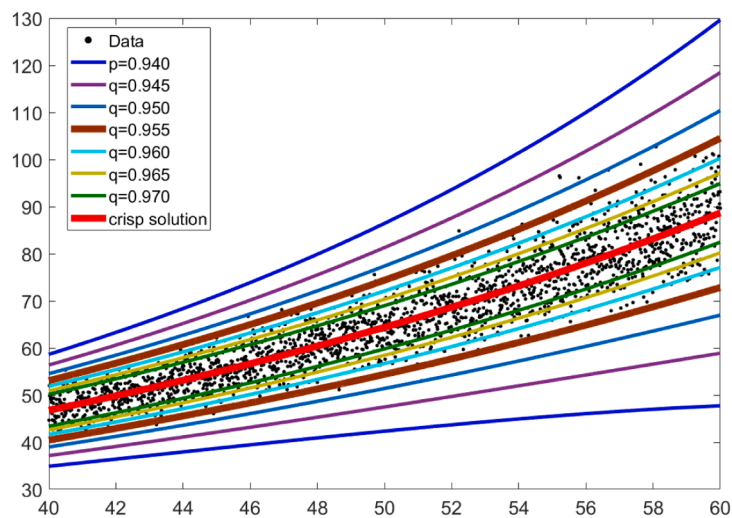


Fig. 8. Simulation results for Dataset 2.

Fig. 9. The change of solutions according to parameter q .

initial condition (20) is

$$\mathbf{S} = \left\{ \mathcal{X}(t) = [x_0 e^{\theta t}]_{\varphi(t)} \mid \varphi : [0, \infty) \rightarrow \mathbb{E}_L^I, \varphi \text{ is continuous, } \varphi(t) \geq_S \mathcal{E}, \text{ and } \varphi(0) = \mathcal{E}^* \right\}.$$

4.2.2. Simulation analysis

We generate two noisy datasets for the population growth problem:

Dataset 1: having the size $N = 2000$, generated by

$$D_1(t) = 13e^{0.032t} + \epsilon$$

at $t = \frac{60}{1999}i$, with $i = 0, 1, \dots, 1999$, where $\epsilon \sim N(0, 1)$.⁴

Dataset 2: having the size $N = 2000$, generated by

$$D_2(t) = (13 + \epsilon_1)e^{(0.032 + \epsilon_2)t}$$

at $t = \frac{60}{1999}i$, with $i = 0, 1, \dots, 1999$, where $\epsilon_1 \sim 0.5 \cdot N(0, 1)$ and $\epsilon_2 \sim 0.001 \cdot N(0, 1)$.

The goal is to use the following problem

$$\begin{cases} \mathcal{X}' = (0.032)_{\mathcal{E}} \mathcal{X} \\ \mathcal{X}(0) = 13_{\mathcal{E}^*} \end{cases}, \quad (21)$$

where the unknown function $\mathcal{X}$ values in $\mathbb{F}_L^I$, for $I = (0, \infty]$, $\mathcal{E} = \mathcal{E}_{400, L}$, and $\mathcal{E}^* = \mathcal{E}_{0.5, L}$, to identify a fuzzy model that fits the two datasets above. As analysis in the previous subsection, the set of solutions of (21) is as follows:

$$\mathbf{S} = \left\{ \mathcal{X}(t) = [13e^{0.032t}]_{\varphi(t)} \mid \varphi : [0, \infty) \rightarrow \mathbb{E}_L^I, \varphi \text{ is continuous, } \varphi(t) \geq_S \mathcal{E}_{400, L}, \text{ and } \varphi(0) = \mathcal{E}_{0.5, L} \right\}.$$

It follows that we need to determine a suitable function φ such that the solution defined with it fits to each given dataset. Known that φ is determined with respect to a function $\beta(t) : [0, \infty) \rightarrow (0, \infty)$ by $\varphi(t) = \mathcal{E}_{\beta(t), L}$. We identify solutions to each dataset based on the calibrating of function φ according to function β of the form $\beta(t) = \alpha q^t$, $\alpha, q > 0$, with the note that a decrease in β leads to an increase in the width of φ , and vice versa. Since, the condition that $\varphi(0) = \mathcal{E}_{0.5, L}$, we obtain $\alpha = 0.5$. Consequently, the found solutions depend on the parameter q , only. The obtained results are as follows:

For Dataset 1, since the uncertainty of the data is unchanged on time, it is reasonable to set $q = 1$ to obtain a solution that is suitable with the dataset. The solution is depicted in Fig. 7b. A zoomed-in version of the solution to the interval $0 \leq t \leq 10$ is displayed in Fig. 7c.

For Dataset 2, since the uncertainty of the data is increasing on time which means that in the found solution the function β must be decreasing on time, it is reasonable to set $q < 1$. By calibrating parameter q , we receive a solution as in Fig. 8b with $q = 0.955$. A comparison of it with the data is shown in the Fig. 8c.

As we can see above, the found fuzzy models for considered datasets strongly depend on the calibration of parameter q , which is a positive real number. For the first dataset, it is easy to see that q should be 1. However, for the second case, the found solution is subjectively obtained by mainly on descriptive statistics. Remind that we can estimate q for the solution using statistical tools, as mentioned in Section 4.1, those are planned for the future studies. In the current paper, we consider the sensitivity of the solutions of (21) according to parameter q . A numerical result in comparison with Dataset 2 is depicted in Fig. 9. Theoretically, if we denote by $\mathcal{X}_q$ the solution of (21) obtain with respect to q , then for any $q > 0$ and a sequence $(q_n)_n$, $q_n > 0$, that converges to q , we obtain that

$$\lim_{q_n \rightarrow q} \mathcal{X}_{q_n}(t) = \mathcal{X}_q(t),$$

for any time t . This is a straightforward consequence of the results in Example 4 under the fact that the function $q \mapsto \alpha q^t$ is continuous for each t .

5. Conclusions

Motivated by the computational simplicity and mathematical convenience in performing arithmetic operations on extensional fuzzy numbers, the operations that can be calculated similarly to those on real numbers (do not stick to the level sets of fuzzy numbers), we took the next step to explore calculus concepts within this set of specific fuzzy numbers and demonstrated their valuable contributions to the theory and application of fuzzy differential equations. We introduced the concepts and explored essential properties of the limit, continuity, derivative, and integral of functions, taking values as extensional fuzzy numbers. We studied the fuzzy initial value problems under the novel framework. Numerical and practical examples were considered to demonstrate novel contributions of the proposed calculus to the theory.

The contributions of the proposed calculus are quite clear reflected in its computational simplicity and its effective applicability to fuzzy initial value problems as analyzed in Sections 2–4. Nevertheless, the calculus has a notable limitation in which it is developed only for the class of extensional fuzzy numbers that are symmetric. This is understandable because extensional fuzzy numbers are

as $10\epsilon_1$ and $10000\epsilon_2$ leads to the fact that $\mathcal{E}_2 \geq_S \mathcal{E}_1$. It means we can convert the counting unit to make the fuzzy similarity $\mathcal{E}_0$ satisfy the mentioned assumption.

⁴ ϵ_1 follows the normal distribution with mean 0 and variance 1.

defined according to fuzzy similarity relations that are symmetric. Although symmetric fuzzy numbers are suitable in several practical situations where the necessity of modeling phenomena using asymmetric fuzzy numbers is unclear, extending the proposed calculus to more general classes is a natural and justified requirement. It is possible to make such a generalization. Indeed, the symmetry property that is a consequence of the use of fuzzy similarity relations for building fuzzy numbers is not essential in the proposed calculus. Whenever a more general definition of fuzzy numbers without using fuzzy similarity relations is devoted, a generalization of the proposed calculus becomes entirely possible. We planned to share our investigations on such a generalization in the near future.

CRedit authorship contribution statement

Linh Nguyen: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization; **Tri Truong:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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